

Adalimumab - Medical Clinical Policy Bulletins

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Table Of Contents

[Policy](#)

[Applicable CPT / HCPCS / ICD-10 Codes](#)

[Background](#)

[References](#)

Policy

Scope of Policy

This Clinical Policy Bulletin addresses the following adalimumab products for commercial medical plans:

1. adalimumab (Humira or unbranded Humira)
2. adalimumab-aacf (Idacio or unbranded Idacio)
3. adalimumab-aaty (Yuflyma or unbranded Yuflyma)
4. adalimumab-adaz (Hyrimoz or unbranded Hyrimoz)
5. adalimumab-adbm (Cyltezo or unbranded Cyltezo)
6. adalimumab-afzb (Abrilada)
7. adalimumab-aqvh (Yusimry)
8. adalimumab-atto (Amjevita)
9. adalimumab-bwwd (Hadlima or unbranded Hadlima)
10. adalimumab-fkjp (Hulio or unbranded Hulio)
11. adalimumab-ryvk (Simlandi or unbranded Simlandi).

1. Prescriber Specialties

This medication must be prescribed by or in consultation with *one* of the following:

1. Rheumatoid arthritis, articular juvenile idiopathic arthritis, ankylosing spondylitis, non-radiographic axial spondyloarthritis, and Behcet's disease: rheumatologist;
2. Psoriatic arthritis and hidradenitis suppurativa: rheumatologist or dermatologist;
3. Crohn's disease and ulcerative colitis: gastroenterologist;
4. Plaque psoriasis and pyoderma gangrenosum: dermatologist;
5. Uveitis: ophthalmologist or rheumatologist;
6. Immune checkpoint inhibitor-related toxicity: oncologist, hematologist, or

rheumatologist.

2. Criteria for Initial Approval

Aetna considers adalimumab, or its biosimilar, medically necessary for the following indications when criteria are met:

1. Rheumatoid arthritis (RA)

1. For adult members who have previously received a biologic or targeted synthetic drug (e.g., Rinvoq, Xeljanz) indicated for moderately to severely active rheumatoid arthritis (RA) within the past 120 days; *or*
2. For adult members for treatment of moderately to severely active RA when *both* of the following criteria are met:
 1. Member meets *either* of the following criteria:
 1. Member has been tested for *either* of the following biomarkers and the test was positive:
 1. Rheumatoid factor (RF); *or*
 2. Anti-cyclic citrullinated peptide (anti-CCP); *or*
 2. Member has been tested for *all* of the following biomarkers:
 1. RF; *and*
 2. Anti-CCP; *and*
 3. C-reactive protein (CRP) and/or erythrocyte sedimentation rate (ESR); *and*
 2. Member meets *one* of the following criteria:
 1. Member has failed to achieve a low disease activity after a 3-month trial of methotrexate (MTX) monotherapy at a maximum titrated dose of at least 15 mg per week and meets *any* of the following conditions:
 1. Member has had a documented inadequate response to MTX in combination with at least one other conventional synthetic drug (i.e., hydroxychloroquine and/or sulfasalazine) after a 3-month trial at a maximum tolerated dose(s); *or*
 2. Member has experienced a documented intolerable adverse event to hydroxychloroquine or sulfasalazine; *or*
 3. Member has a documented contraindication to hydroxychloroquine (see [Appendix A](#)) and sulfasalazine (e.g., porphyria, intestinal or urinary obstruction); *or*
 4. Member has moderate to high disease activity; *or*
 2. Member was unable to tolerate a 3-month trial of MTX monotherapy at a maximum titrated dose of at least 15 mg per week and meets *any* of the following conditions:
 1. Member has had a documented inadequate response to MTX in combination with at least one other conventional synthetic drug (i.e., hydroxychloroquine and/or sulfasalazine)
 2. After a 3-month trial at a maximum tolerated dose(s); *or*
 3. Member has stopped taking MTX and has had a documented inadequate response to another conventional synthetic drug (i.e., leflunomide, hydroxychloroquine, and/or sulfasalazine) alone or in combination after a 3-month trial at a maximum tolerated dose(s); *or*
 4. Member has experienced a documented intolerable adverse event to leflunomide, hydroxychloroquine, or sulfasalazine; *or*
 3. Member has a documented contraindication to leflunomide, hydroxychloroquine (see [Appendix A](#)), and sulfasalazine (e.g., porphyria, intestinal or urinary obstruction); *or*
 6. Member has moderate to high disease activity; *or*
 3. Member has experienced a documented intolerable adverse event or has a documented contraindication to MTX (see [Appendix A](#)), discontinues MTX, and meets *any* of the following conditions:

1. Member has had a documented inadequate response to another conventional synthetic drug (i.e., leflunomide, hydroxychloroquine, and/or sulfasalazine) alone or in combination
2. after a 3-month trial at a maximum tolerated dose(s); *or*
3. Member has experienced a documented intolerable adverse event to leflunomide, hydroxychloroquine, or sulfasalazine; *or*
4. Member has a documented contraindication to leflunomide, hydroxychloroquine (see [Appendix A](#)), and sulfasalazine (e.g., porphyria, intestinal or urinary obstruction); *or*
5. Member has moderate to high disease activity;

2. Articular juvenile idiopathic arthritis (JIA)

1. For members 2 years of age or older who have previously received a biologic or targeted synthetic drug (e.g., Xeljanz) indicated for moderately to severely active articular juvenile idiopathic arthritis; *or*
2. For members 2 years of age or older for treatment of moderate to severely active articular JIA when *any* of the following criteria is met:
 1. Member has had an inadequate response to methotrexate or another conventional synthetic drug (e.g., leflunomide, sulfasalazine, hydroxychloroquine) administered at an adequate dose and duration; *or*
 2. Member has had an inadequate response to a trial of scheduled non-steroidal anti-inflammatory drugs (NSAIDs) and/or intra-articular glucocorticoids (e.g., triamcinolone hexacetonide) and *one* of the following risk factors for poor outcome:
 1. Involvement of ankle, wrist, hip, sacroiliac joint, and/or temporomandibular joint (TMJ); *or*
 2. Presence of erosive disease or enthesitis; *or*
 3. Delay in diagnosis; *or*
 4. Elevated levels of inflammation markers; *or*
 5. Symmetric disease;
3. Member has risk factors for disease severity and potentially a more refractory disease course (see [Appendix B](#)) and the member also meets *one* of the following:
 1. High-risk joints are involved (e.g., cervical spine, wrist, or hip); *or*
 2. High disease activity; *or*
 3. Is judged to be at high risk for disabling joint disease;

3. Psoriatic arthritis (PsA)

1. For adult members who have previously received a biologic or targeted synthetic drug (e.g., Rinvoq, Otezla) indicated for active psoriatic arthritis; *or*
2. For adult members for treatment of active psoriatic arthritis when *either* of the following criteria is met:
 1. Member has mild to moderate disease and meets *one* of the following criteria:
 1. Member has had an inadequate response to methotrexate, leflunomide, or another conventional synthetic drug (e.g., sulfasalazine) administered at an adequate dose and duration; *or*
 2. Member has an intolerance or contraindication to methotrexate or leflunomide (see [Appendix A](#)), or another conventional synthetic drug (e.g., sulfasalazine); *or*
 3. Member has enthesitis or predominantly axial disease; *or*
 2. Member has severe disease;

4. Ankylosing spondylitis (AS) and non-radiographic axial spondyloarthritis (nr-axSpA)

1. For adult members who have previously received a biologic or targeted synthetic drug

(e.g., Rinvoq, Xeljanz) indicated for active ankylosing spondylitis or active non-radiographic axial spondyloarthritis; *or*

2. For adult members for treatment of active ankylosing spondylitis or active non-radiographic axial spondyloarthritis when *either* of the following criteria is met:
 1. Member has had an inadequate response to at least two non-steroidal anti-inflammatory drugs (NSAIDs); *or*

2. Member has an intolerance or contraindication to two or more NSAIDs;

5. Crohn's disease (CD)

For treatment of moderately to severely active Crohn's disease;

6. Ulcerative colitis (UC)

For treatment of moderately to severely active ulcerative colitis;

7. Plaque psoriasis (PsO)

1. For adult members who have previously received a biologic or targeted synthetic drug (e.g., Sotyktu, Otezla) indicated for the treatment of moderate to severe plaque psoriasis; *or*
2. For adult members for treatment of moderate to severe plaque psoriasis when *any* of the following criteria is met:
 1. Crucial body areas (e.g., hands, feet, face, neck, scalp, genitals/groin, intertriginous areas) are affected; *or*
 2. At least 10% of the body surface area (BSA) is affected; *or*
 3. At least 3% of body surface area (BSA) is affected and the member meets *any* of the following criteria:
 1. Member has had an inadequate response or intolerance to either phototherapy (e.g., UVB, PUVA) or pharmacologic treatment with methotrexate, cyclosporine, or acitretin; *or*
 2. Member has a clinical reason to avoid pharmacologic treatment with methotrexate, cyclosporine and acitretin (see [Appendix A](#));

8. Hidradenitis suppurativa

1. For members 12 years of age or older who have previously received a biologic indicated for the treatment of moderate to severe hidradenitis suppurativa; *or*
2. For members 12 years of age or older for treatment of moderate to severe hidradenitis suppurativa when *either* of the following is met:
 1. Member has had an inadequate response to an oral antibiotic used for the treatment of hidradenitis suppurativa for at least 90 days (e.g., clindamycin, metronidazole, moxifloxacin, rifampin, tetracyclines); *or*
 2. Member has an intolerance or contraindication to oral antibiotics used for the treatment of hidradenitis suppurativa;

9. Uveitis (non-infectious intermediate, posterior and panuveitis)

1. For members 2 years of age or older who have previously received a biologic indicated for non-infectious intermediate, posterior, and panuveitis; *or*
2. For members 2 years of age or older for treatment of non-infectious intermediate, posterior and panuveitis when *either* of the following is met:
 1. Member has had an inadequate response to corticosteroids or immunosuppressive therapy (e.g., azathioprine, cyclosporine, methotrexate, mycophenolate mofetil); *or*
 2. Member has an intolerance or contraindication to corticosteroids and immunosuppressive therapy (e.g., azathioprine, cyclosporine, methotrexate, mycophenolate mofetil);

10. Behcet's disease

1. For members who have previously received apremilast (Otezla) or a biologic indicated for the treatment of Behçet's disease; *or*
2. For the treatment of Behçet's disease when the member has had an inadequate response to at least one non-biologic medication for Behçet's disease (e.g., azathioprine, colchicine, cyclosporine, systemic corticosteroids);

11. Pyoderma gangrenosum

1. For members who have previously received a biologic indicated for the treatment of pyoderma gangrenosum; *or*
2. For treatment of pyoderma gangrenosum when *either* of the following is met:
 1. Member has had an inadequate response to corticosteroids or immunosuppressive therapy (e.g., cyclosporine or mycophenolate mofetil); *or*
 2. Member has an intolerance or contraindication to corticosteroids and immunosuppressive therapy (e.g. cyclosporine, mycophenolate mofetil);

12. Immune checkpoint inhibitor-related toxicity

For treatment of immune checkpoint inhibitor-related toxicity when the member has moderate or severe immunotherapy-related inflammatory arthritis and meets *either* of the following:

1. Member has had an inadequate response to corticosteroids or a conventional synthetic drug (e.g., methotrexate, sulfasalazine, leflunomide, hydroxychloroquine); *or*
2. Member has an intolerance or contraindication to corticosteroids and a conventional synthetic drug (e.g., methotrexate, sulfasalazine, leflunomide, hydroxychloroquine).

Aetna considers all other indications as experimental, investigational, or unproven.

3. Continuation of Therapy

Aetna considers continuation of adalimumab or its biosimilars medically necessary for the following indications:

1. Rheumatoid arthritis (RA)

For adult all members (including new members) who are using the requested medication for moderately to severely active rheumatoid arthritis and who achieve or maintain a positive clinical response as evidenced by disease activity improvement of at least 20% from baseline in tender joint count, swollen joint count, pain, or disability;

2. Articular juvenile idiopathic arthritis (JIA)

For all members 2 years of age or older (including new members) who are using the requested medication for moderately to severely active articular JIA and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when there is improvement in *any* of the following from baseline:

1. Number of joints with active arthritis (e.g., swelling, pain, limitation of motion); *or*
2. Number of joints with limitation of movement; *or*
3. Functional ability;

3. Psoriatic arthritis (PsA)

For all adult members (including new members) who are using the requested medication for psoriatic arthritis and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when there is improvement in *any* of the following from baseline:

1. Number of swollen joints; *or*

2. Number of tender joints; *or*
3. Dactylitis; *or*
4. Enthesitis; *or*
5. Axial disease; *or*
6. Skin and/or nail involvement; *or*
7. Functional status; *or*
8. C-reactive protein (CRP);

4. Ankylosing spondylitis (AS) and non-radiographic axial spondyloarthritis (nr-axSpA)

For all adult members (including new members) who are using the requested medication for ankylosing spondylitis or non-radiographic axial spondyloarthritis and who achieve or maintain positive clinical response with the requested medication as evidenced by low disease activity or improvement in signs and symptoms of the condition when there is improvement in *any* of the following from baseline:

1. Functional status; *or*
2. Total spinal pain; *or*
3. Inflammation (e.g., morning stiffness); *or*
4. Swollen joints; *or*
5. Tender joints; *or*
6. C-reactive protein (CRP);

5. Crohn's disease (CD)

1. For all members (including new members) who are using the requested medication for moderately to severely active CD and who achieve or maintain remission; *or*
2. For all members (including new members) who are using the requested medication for moderately to severely active CD and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when there is improvement in *any* of the following from baseline:
 1. Abdominal pain or tenderness; *or*
 2. Diarrhea; *or*
 3. Body weight; *or*
 4. Abdominal mass; *or*
 5. Hematocrit; *or*
 6. Appearance of the mucosa on endoscopy, computed tomography enterography (CTE), magnetic resonance enterography (MRE), or intestinal ultrasound; *or*
 7. Improvement on a disease activity scoring tool (e.g., Crohn's Disease Activity Index [CDAI] score);

6. Ulcerative colitis (UC)

1. For all members (including new members) who are using the requested medication for moderately to severely active UC and who achieve or maintain remission; *or*
2. For all members (including new members) who are using the requested medication for moderately to severely active UC and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when there is improvement in *any* of the following from baseline:
 1. Stool frequency; *or*
 2. Rectal bleeding; *or*

3. Urgency of defecation; *or*
4. C-reactive protein (CRP); *or*
5. Fecal calprotectin (FC); *or*
6. Appearance of the mucosa on endoscopy, computed tomography enterography (CT), magnetic resonance enterography (MRE), or intestinal ultrasound; *or*
7. Improvement on a disease activity scoring tool (e.g., Ulcerative Colitis Endoscopic Index of Severity [UCEIS], Mayo score);

7. Plaque psoriasis (PsO)

For all adult members (including new members) who are using the requested medication for moderate to severe plaque psoriasis and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when any of the following is met:

1. Reduction in body surface area (BSA) affected from baseline; *or*
2. Improvement in signs and symptoms from baseline (e.g., itching, redness, flaking, scaling, burning, cracking, or pain);

8. Hidradenitis suppurativa

For all members 12 years of age or older (including new members) who are using the requested medication for moderate to severe hidradenitis suppurativa and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when *any* of the following is met:

1. Reduction in abscess and inflammatory nodule count from baseline; *or*
2. Reduced formation of new sinus tracts and scarring; *or*
3. Decrease in frequency of inflammatory lesions from baseline; *or*
4. Reduction in pain from baseline; *or*
5. Reduction in suppuration from baseline; *or*
6. Improvement in frequency of relapses from baseline; *or*
7. Improvement in quality of life from baseline; *or*
8. Improvement on a disease severity assessment tool from baseline;

9. Uveitis (non-infectious intermediate, posterior and panuveitis)

For all members 2 years of age or older (including new members) who are using the requested medication for non-infectious intermediate, posterior, and panuveitis and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when the patient meets *any* of the following:

1. Reduced frequency of disease flares compared to baseline; *or*
2. Stability or improvement in anterior chamber (AC) cell grade compared to baseline; *or*
3. Stability or improvement in vitreous haze (VH) grade compared to baseline; *or*
4. Stability or improvement in visual acuity compared to baseline; *or*
5. Reduction in glucocorticoid requirements from baseline; *or*
6. No new active inflammatory chorioretinal and/or inflammatory retinal vascular lesions relative to baseline;

10. Immune checkpoint inhibitor-related toxicity

For all members (including new members) who are using the requested medication for

immunotherapy-related inflammatory arthritis and who achieve or maintain a positive clinical response with the requested medication as evidenced by low disease activity or improvement in signs and symptoms of the condition.

11. All other indications

For all members (including new members) who are using the requested medication for an indication outlined in Section II and who achieve or maintain a positive clinical response with the requested medication as evidenced by low disease activity or improvement in signs and symptoms of the condition.

4. Other

For all indications: Member has had a documented negative tuberculosis (TB) test (which can include a tuberculosis skin test [TST] or an interferon-release assay [IGRA])^{*} within 12 months of initiating therapy for persons who are naïve to biologic drugs or targeted synthetic drugs associated with an increased risk of TB.

* If the screening testing for TB is positive, there must be further testing to confirm there is no active disease (e.g., chest x-ray). Do not administer the requested medication to members with active TB infection. If there is latent disease, TB treatment must be started before initiation of the requested medication.

For all indications: Member cannot use the requested medication concomitantly with any other biologic drug or targeted synthetic drug for the same indication.

5. Related Policies

1. [CPB 0205 - Phototherapy and Photochemotherapy \(PUVA\) for Skin Conditions](#)
2. [CPB 0314 - Rituximab](#)
3. [CPB 0315 - Etanercept](#)
4. [CPB 0341 - Infliximab](#)
5. [CPB 0577 - Laser Treatment for Psoriasis and Other Selected Skin Conditions](#)
6. [CPB 0720 - Abatacept \(Orencia\)](#)
7. [CPB 0761 - Certolizumab Pegol \(Cimzia\)](#)
8. [CPB 0790 - Golimumab \(Simponi and Simponi Aria\)](#)
9. [CPB 0905 - Secukinumab \(Cosentyx\)](#)
10. [CPB 0912 - Ustekinumab](#)
11. [CPB 1009 - Risankizumab-rzaa \(Skyrizi\)](#)
12. [CPB 1011 - Guselkumab \(Tremfya\)](#)
13. [CPB 1091 - Therapeutic Drug Monitoring of Anti-Tumor Necrosis Factor Blocking Agents](#) - for measurement of serum levels of adalimumab and/or antibodies to adalimumab

Dosage and Administration

Note: Approvals may be subject to dosing limits in accordance with FDA-approved labeling, accepted compendia, and/or evidence-based practice guidelines. For rheumatoid arthritis, member must initiate treatment with every other week dosing. Below includes dosing recommendations as per the FDA-approved prescribing information.

Note: Adalimumab is available in generic and biosimilar formulations. Some of the branded (proprietary name) biosimilar products below may be available in an unbranded equivalent. Furthermore, some adalimumab biosimilars have received FDA-approval for interchangeable designation. Interchangeability designations allow for a biosimilar to be substituted for a reference product at the pharmacy level without having to obtain permission from a physician, depending on state pharmacy laws. This is similar to how generic drugs are routinely substituted for brand-name drugs. Not all biosimilars are interchangeable.

Humira

Humira is supplied as the following for subcutaneous injection:

- Single-dose prefilled pen (Humira Pen): 80 mg/0.8 mL, 40 mg/0.8 mL, and 40 mg/0.4 mL
- Single-dose prefilled glass syringe: 80 mg/0.8 mL, 40 mg/0.8 mL, 40 mg/0.4 mL, 20 mg/0.4 mL, 20 mg/0.2 mL, 10 mg/0.2 mL, 10 mg/0.1 mL
- Single-dose glass vial for institutional use only: 40 mg/0.8 mL.

Rheumatoid Arthritis, Psoriatic Arthritis, and Ankylosing Spondylitis

The recommended subcutaneous dose of Humira for adults with rheumatoid arthritis (RA), psoriatic arthritis (PsA), or ankylosing spondylitis (AS) is 40 mg administered every other week. Methotrexate (MTX), other non-biologic DMARDs, glucocorticoids, nonsteroidal anti-inflammatory drugs (NSAIDs), and/or analgesics may be continued during treatment with Humira. In the treatment of RA, some individuals not taking concomitant MTX may derive additional benefit from increasing the dosing frequency of Humira to 40 mg every week or 80 mg every other week.

Juvenile Idiopathic Arthritis or Pediatric Uveitis

The recommended subcutaneous dosage of Humira for persons 2 years of age and older with polyarticular juvenile idiopathic arthritis (JIA) or pediatric uveitis is based on weight shown below. MTX, glucocorticoids, NSAIDs, and/or analgesics may be continued during treatment with Humira.

Table: Recommended Dose of Humira for Persons 2 Years of Age and Older with Polyarticular Juvenile Idiopathic Arthritis (JIA) or Pediatric Uveitis

Pediatric Weight (2 years of age and older)	Recommended Dose
10 kg (22 lbs) to less than 15 kg (33 lbs)	10 mg every other week
15 kg (33 lbs) to less than 30 kg (66 lbs)	20 mg every other week
30 kg (66 lbs) and greater	40 mg every other week

Humira has not been studied in persons with polyarticular JIA or pediatric uveitis less than 2 years of age or in persons with a weight below 10 kg.

Crohn's Disease

Adults:

The recommended subcutaneous Humira dose regimen for adults with Crohn's disease (CD) is 160 mg initially on Day 1 (given in one day or split over two consecutive days), followed by 80 mg two weeks later (Day 15). Two weeks later (Day 29) begin a maintenance dose of 40 mg every other week. Aminosalicylates and/or corticosteroids may be continued during treatment with Humira. Azathioprine, 6-mercaptopurine (6-MP) or MTX may be continued during treatment with Humira if necessary.

Pediatrics:

The recommended subcutaneous Humira dose regimen for pediatrics 6 years of age and older with Crohn's disease (CD) is based on body weight as shown below:

Table: Recommended Humira Dose Regimen for Pediatrics 6 Years of Age and Older with Crohn's Disease (CD) Based on Body Weight

Pediatric Weight	Days 1 through 15	Starting on Day 29
17 kg (37 lbs) to less than 40 kg (88 lbs)	Day 1: 80 mg	20 mg every other week
	Day 15: 40 mg	

40 kg (88 lbs) and greater	Day 1: 160 mg (single dose or split over two consecutive days)	40 mg every other week
	Day 15: 80 mg	

Ulcerative Colitis (UC)

Adults:

The recommended subcutaneous Humira dose regimen for adults with UC is 160 mg initially on Day 1 (given in one day or split over two consecutive days), followed by 80 mg two weeks later (Day 15). Two weeks later (Day 29) continue with a dose of 40 mg every other week.

Only continue Humira in persons who have shown evidence of clinical remission by eight weeks (Day 57) of therapy. Aminosalicylates and/or corticosteroids may be continued during treatment with Humira. Azathioprine and 6-mercaptopurine (6-MP) may be continued during treatment with Humira if necessary.

Pediatrics:

The recommended subcutaneous dosage of Humira for pediatrics 5 years of age and older with UC is based on body weight as shown below:

Table: Recommended Humira Dose Regimen for Pediatrics 5 Years of Age and Older with Ulcerative Colitis (CD) Based on Body Weight

Pediatric Weight	Days 1 through 15	Starting on Day 29**
20 kg (44 lbs) to less than 40 kg (88 lbs)	Day 1: 80 mg	40 mg every other week or 20 mg every week
	Day 8: 40 mg	
	Day 15: 40 mg	
40 kg (88 lbs) and greater	Day 1: 160 mg (single dose or split over two consecutive days)	80 mg every other week or 40 mg every week
	Day 8: 80 mg	
	Day 15: 80 mg	

** Continue the recommended pediatric dosage in persons with UC who turn 18 years of age and who are well-controlled on their Humira regimen.

Plaque Psoriasis or Adult Uveitis

The recommended subcutaneous dose of Humira for adults with plaque psoriasis or uveitis is an initial dose of 80 mg, followed by 40 mg given every other week starting one week after the initial dose. The use of Humira in moderate to severe chronic plaque psoriasis beyond one year has not been evaluated in controlled clinical studies.

Hidradenitis Suppurativa

Adults:

The recommended subcutaneous dose of Humira for adults with hidradenitis suppurativa (HS) is an initial dose of 160 mg (given in one day or split over two consecutive days), followed by 80 mg two weeks later (Day 15). Begin 40 mg weekly or 80 mg every other week dosing two weeks later (Day 29).

Adolescents:

The recommended subcutaneous dose of Humira for adolescents 12 years of age and older

weighing at least 30 kg with HS is based on body weight as shown below:

Table: Recommended Dose of Humira for Adolescents 12 Years of Age and Older Weighing at Least 30 kg with Hidradenitis Suppurativa (HS) Based on Body Weight

Body Weight of Adolescents (12 years of age and older)	Recommended Dosage Regimen
30 kg (66 lbs) to less than 60 kg (132 lbs)	Day 1: 80 mg
	Day 8 and subsequent doses: 40 mg every other week
60 kg (132 lbs) and greater	Day 1: 160 mg (given in one day or split over two consecutive days)
	Day 15: 80 mg
	Day 29 and subsequent doses: 40 mg every week or 80 mg every other week

Source: AbbVie, 2024

Abrilada

Abrilada (adalimumab-afzb) is considered a biosimilar product to Humira.

Abrilada is supplied as the following for subcutaneous injection:

- Single-dose prefilled pen (ABRILADA Pen): 40 mg/0.8 mL
- Single-dose prefilled glass syringe: 40 mg/0.8 mL, 20 mg/0.4 mL, 10 mg/0.2 mL
- Single-dose glass vial for institutional use only: 40 mg/0.8 mL.

Abrilada carries the same labeled indications as Humira except for adolescent hidradenitis suppurativa and pediatric ulcerative colitis.

Source: Pfizer, 2024

Amjevita

Amjevita (adalimumab-atto) is considered a biosimilar product to Humira. It is available for subcutaneous injection and supplied as the following:

- Single-dose prefilled SureClick® autoinjector: 40 mg/0.4 mL, 40 mg/0.8 mL, 80 mg/0.8 mL
- Single-dose prefilled glass syringe: 10 mg/0.2 mL, 20 mg/0.2 mL, 20 mg/0.4 mL, 40 mg/0.4 mL, 40 mg/0.8 mL, 80 mg/0.8 mL

Amjevita carries the same labeled indications as Humira except for adolescent hidradenitis suppurativa and pediatric ulcerative colitis.

Source: Amgen, 2023

Cyltezo

Cyltezo (adalimumab-adbm) is considered a biosimilar product to Humira. It is available for subcutaneous injection and supplied as the following:

- Single-dose prefilled pen (Cyltezo Pen): 40 mg/0.4 mL, 40 mg/0.8 mL
- Single-dose prefilled glass syringe: 10 mg/0.2 mL, 20 mg/0.4 mL, 40 mg/0.4 mL, 40 mg/0.8 mL

Cyltezo carries the same labeled indications as Humira except for adolescent hidradenitis suppurativa and pediatric ulcerative colitis.

Note: adalimumab-adbm is also available as an unbranded generic.

Source: Boehringer Ingelheim, 2024

Hadlima

Hadlima (adalimumab-bwvd) is considered a biosimilar to Humira. It is available for subcutaneous injection and supplied as the following:

- Single-dose prefilled autoinjector (HADLIMA PushTouch): 40 mg/0.4 mL, 40 mg/0.8 mL
- Single-dose prefilled glass syringe: 40 mg/0.4 mL, 40 mg/0.8 mL
- Single-dose glass vial for institutional use only: 40 mg/0.8 mL

Hadlima carries the same labeled indications as Humira except for adolescent hidradenitis suppurativa and pediatric ulcerative colitis.

Source: Organon & Co, 2024

Hulio

Hulio (adalimumab-fkjp) is considered a biosimilar to Humira. It is available for subcutaneous injection and supplied as the following:

- Injection: 40 mg/0.8 mL in a single-dose prefilled pen (Hulio Pen)
- Injection: 40 mg/0.8 mL in a single-dose prefilled plastic syringe
- Injection: 20 mg/0.4 mL in a single-dose prefilled plastic syringe.

Hulio carries the same labeled indications as Humira except for adolescent hidradenitis suppurativa and pediatric ulcerative colitis.

Note: adalimumab-fkjp is also available as an unbranded generic.

Source: Biocon Biologics, 2023

Hyrimoz

Hyrimoz (adalimumab-adaz) is considered a biosimilar to Humira. It is available for subcutaneous injection and supplied as the following:

- Single-dose prefilled pen (Sensoready Pen): 40 mg/0.8 mL, 40 mg/0.4 mL and 80 mg/0.8 mL
- Single-dose prefilled glass syringe (with BD UltraSafe Passive™ Needle Guard): 20 mg/0.4 mL, 40 mg/0.8 mL, 40 mg/0.4 mL and 80 mg/0.8 mL
- Single-dose prefilled glass syringe: 10 mg/0.2 mL, 10 mg/0.1 mL and 20 mg/0.2 mL

Hyrimoz carries the same labeled indications as Humira except for adolescent hidradenitis suppurativa, and pediatric ulcerative colitis.

Note: adalimumab-adaz is also available as an unbranded generic.

Source: Sandoz Inc, 2024

Idacio

Idacio (adalimumab-aacf) is considered a biosimilar to Humira. It is available for subcutaneous injection and supplied as the following:

- Single-dose prefilled pen (IDACIO Pen): 40 mg/0.8 mL
- Single-dose prefilled glass syringe: 40 mg/0.8 mL
- Single dose glass vial kit for institutional use only: 40mg/0.8 mL

Idacio carries the same labeled indications as Humira except for adolescent hidradenitis suppurativa and pediatric ulcerative colitis.

Note: adalimumab-aacf is also available as an unbranded generic.

Source: Fresenius Kabi USA, 2024

Simlandi

Simlandi (adalimumab-ryvk) is considered a biosimilar to Humira. It is available for subcutaneous injection and supplied as a 40 mg/0.4 mL single-dose autoinjector.

Simlandi carries the same labeled indications as Humira except for adolescent hidradenitis suppurativa, pediatric ulcerative colitis, and pediatric uveitis.

Source: Alvotect USA Inc., 2024

Yuflyma

Yuflyma (adalimumab-aaty) is considered a biosimilar to Humira. It is available for subcutaneous injection and supplied as the following:

- Single-dose prefilled auto-injector (Yuflyma AI): 40 mg/0.4 mL, 80 mg/0.8 mL
- Single-dose prefilled syringe with safety guard : 40 mg/0.4 mL, 80 mg/0.8 mL
- Single-dose prefilled syringe: 20 mg/0.2 mL, 40 mg/0.4 mL, 80 mg/0.8 mL

Yuflyma carries the same labeled indications as Humira except for adolescent hidradenitis suppurativa and pediatric ulcerative colitis.

Note: adalimumab-aaty is also available as an unbranded generic.

Source: Celltrion USA, 2024

Yusimry

Yusimry (adalimumab-aqvh) is considered a biosimilar to Humira. It is available for subcutaneous injection and supplied as the following:

- Single-dose prefilled pen (YUSIMRY Pen): 40 mg/0.8 mL
- Single-dose prefilled glass syringe: 40 mg/0.8 mL

Yusimry carries the same labeled indications as Humira except for adolescent hidradenitis suppurativa and pediatric ulcerative colitis.

Source: Meitheal Pharmaceuticals, 2024

Experimental, Investigational, or Unproven

1. Aetna considers concomitant use of adalimumab with any other biologic drug (e.g., abatacept, anakinra, etanercept, infliximab, tocilizumab) or targeted synthetic drug (e.g. tofacitinib) experimental, investigational, or unproven for the same indication because the effectiveness of this approach has not been established.
2. Aetna considers adalimumab experimental, investigational, or unproven for all other indications, including any of the following conditions (not an all-inclusive list), because the safety and effectiveness of adalimumab for these conditions has not been established:
 1. Active infections; *or*
 2. Alopecia areata; *or*
 3. Asthma; *or*
 4. Cellulitis; *or*
 5. Chronic recurrent multifocal osteomyelitis (CRMO); *or*
 6. Cryopyrin-associated periodic syndrome (CAPS); *or*
 7. Cystoid macular degeneration; *or*
 8. Dissecting scalp cellulitis, *or*
 9. Giant-cell arteritis; *or*
 10. Granuloma annulare (also known as necrobiotic papulosis); *or*
 11. Guttate psoriasis; *or*

12. Hunter syndrome (mucopolysaccharidosis type II); *or*
 13. Hurler syndrome (mucopolysaccharidosis type I); *or*
 14. Infectious uveitis; *or*
 15. Inflammatory polyarthropathy; *or*
 16. Lichen planus; *or*
 17. Lupus pernio; *or*
 18. Osteoarthritis; *or*
 19. Recurrent pregnancy loss; *or*
 20. Reiter's syndrome (reactive arthritis); *or*
 21. Relapsing polychondritis; *or*
 22. Sarcoidosis; *or*
 23. Sciatica; *or*
 24. Takayasu's arteritis; *or*
 25. Vogt-Koyanagi syndrome.
-

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
Other CPT codes related to CPB:	
71045 - 71048	Radiologic examination, chest
85651	Sedimentation rate, erythrocyte; non-automated
85652	Sedimentation rate, erythrocyte; automated
86140	C-reactive protein
86141	C-reactive protein; high sensitivity (hsCRP)
86200	Cyclic citrullinated peptide (CCP), antibody
86430	Rheumatoid factor; qualitative
86431	Rheumatoid factor; quantitative
86480	Tuberculosis test, cell mediated immunity antigen response measurement; gamma interferon
86481	Tuberculosis test, cell mediated immunity antigen response measurement; enumeration of gamma interferon - producing T cells in cell suspension
86580	Skin test; tuberculosis, intradermal
96372	Therapeutic, prophylactic, or diagnostic injection (specify substance or drug); subcutaneous or intramuscular
96401 - 96450	Chemotherapy administration

Code	Code Description
HCPCS codes covered if selection criteria are met:	
<i>adalimumab-atto (Amjevita), adalimumab-bwwd (unbranded Hadlima), adalimumab-adaz (Hyrimoz), adalimumab-aqvh (Yusimry), adalimumab-ryvk (Simlandi)-no specific code</i>	
J0139	Injection, adalimumab, 1 mg
Q5140	Injection, adalimumab-fkjp, biosimilar, 1 mg
Q5141	Injection, adalimumab-aaty, biosimilar, 1 mg
Q5142	Injection, adalimumab-aaty, biosimilar, 1 mg
Q5143	Injection, adalimumab-adbm, biosimilar, 1 mg
Q5144	Injection, adalimumab-aacf (idacio), biosimilar, 1 mg
Q5145	Injection, adalimumab-afzb (abrilada), biosimilar, 1 mg
Other HCPCS codes related to the CPB:	
<i>Anakinra (Kineret), Rinvoq, Otezla and Xeljanz, Clindamycin, Metronidazole, Rifampicin, Sulfasalazine, Leflunomide - - no specific code:</i>	
J0120	Injection, tetracycline, up to 250 mg
J0129	Injection, abatacept, 10 mg (code may be used for medicare when drug administered under the direct supervision of a physician, not for use when drug is self-administered)
J0702	Injection, betamethasone acetate 3 mg and betamethasone sodium phosphate 3 mg [Celestone]
J1094	Injection, dexamethasone acetate, 1 mg
J1100	Injection, dexamethasone sodium phosphate, 1 mg
J1130	Injection, diclofenac sodium, 0.5 mg
J1438	Injection, etanercept, 25 mg (code may be used for Medicare when drug administered under the direct supervision of a physician, not for use when drug is self-administered)
J1600	Injection, gold sodium thiolamate, up to 50 mg
J1720	Injection, hydrocortisone sodium succinate, up to 100 mg [Cortef]
J1741	Injection, ibuprofen, 100 mg
J1745	Injection, infliximab, 10 mg
J1885	Injection, ketorolac tromethamine, per 15 mg
J2650	Injection, prednisolone acetate, up to 1 ml
J2280	Injection, moxifloxacin, 100 mg
J2281	Injection, moxifloxacin (fresenius kabi) not therapeutically equivalent to j2280, 100 mg

Code	Code Description
J2804	Injection, rifampin, 1 mg
J3245	Injection, tildrakizumab, 1 mg
J3300 - J3301	Injection, triamcinolone acetoneide
J3302	Injection, triamcinolone diacetate, per 5 mg
J3303	Injection, triamcinolone hexacetoneide, per 5 mg
J7500	Azathioprine, oral, 50 mg
J7501	Azathioprine, parenteral, 100 mg
J7502	Cyclosporine, oral, 100mg
J7510	Prednisolone, oral, per 5 mg [Cortef]
J7512	Prednisone, immediate release or delayed release, oral, 1 mg
J7515	Cyclosporine, oral, 25 mg
J7516	Cyclosporine, parenteral, 250 mg
J8530	Cyclophosphamide; oral, 25 mg
J8540	Injection, dexamethasone acetate, 1 mg
J8610	Methotrexate; oral, 2.5 mg
J8611	Methotrexate (jylamvo), oral, 2.5 mg
J8612	Methotrexate (xatmep), oral, 2.5 mg
J9070	Cyclophosphamide, 100 mg
J9073	Injection, cyclophosphamide (ingenus), 5 mg
J9074	Injection, cyclophosphamide (sandoz), 5 mg
J9075	Injection, cyclophosphamide, not otherwise specified, 5mg
J9076	Injection, cyclophosphamide (baxter), 5 mg
J9250	Methotrexate sodium, 5 mg
J9255	Injection, methotrexate (accord) not therapeutically equivalent to j9250 or j9260, 50 mg
J9260	Methotrexate sodium, 50 mg
Q5109	Injection, infliximab-qbtx, biosimilar, (ixifi), 10 mg
S9359	Home infusion therapy, anti-tumor necrosis factor intravenous therapy; (e.g., Infliximab); administrative services, professional pharmacy services, care coordination, and all necessary supplies and equipment (drugs and nursing visits coded separately), per diem
ICD-10 codes covered if selection criteria are met:	
H20.051 - H20.9	Unspecified iridocyclitis [uveitis] [age 2 and older]

Code	Code Description
H44.111 - H44.119	Panuveitis [age 2 and older]
H44.131 - H44.139	Sympathetic uveitis [age 2 and older]
K50.00 - K50.919	Crohn's disease [regional enteritis] [active Crohn's disease with listed manifestations]
K51.00 - K51.919	Ulcerative colitis
• L40.0 - L04.3, • L40.50 - L40.9	Psoriasis [age 18 and older]
L73.2	Hidradenitis suppurativa [age 12 and older]
L88	Pyoderma gangrenosum
• M05.00 - M05.09, • M05.20 - M06.39, • M06.80 - M06.9	Rheumatoid arthritis [moderately to severely active in adults] [age 18 and older]
M08.00 - M08.99	Juvenile idiopathic arthritis [age 2 and older]
M35.2	Behcet's disease
M45.0 - M45.AB	Ankylosing spondylitis [active and failed two or more NSAIDS] [age 18 and older]
T45.AX5A - T45.AX5S	Adverse effect of immune checkpoint inhibitors and immunostimulant drugs [inflammatory arthritis]
ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):	
A00.0 - B99.9	Infectious and parasitic diseases [active]
E76.01 - E76.03	Mucopolysaccharidosis, type I
E76.1	Mucopolysaccharidosis, type II [Hunter's syndrome]
H05.011 - H05.019	Cellulitis of orbit
• H20.00 - H20.039, • H20.05 - H20.9	Acute and subacute iridocyclitis, chronic iridocyclitis, certain types of iridocyclitis, and unspecified iridocyclitis [other than noninfectious]
H30.001 - H30.93	Chorioretinal inflammation

Code	Code Description
H35.351 - H35.359	Cystoid macular degeneration
J45.20 - J45.998	Asthma
K12.2	Cellulitis and abscess of mouth
• L02.01, • L02.11, • L02.211 - L02.219 • L02.31, • L02.411 - L02.419 • L02.511 - L02.519, • L02.611 - L02.619 • L02.811 - L02.818, • L02.91 • L03.111 - L03.91	Other cellulitis and abscess
L03.011 - L03.049	Cellulitis and abscess of finger and toe
L03.811	Cellulitis of head [any part] [dissecting scalp cellulitis]
L40.4	Guttate psoriasis
L43.0 - L43.9	Lichen planus
L63.0 - L63.9	Alopecia areata
L92.0	Granuloma annulare
M02.30 - M02.39	Reiter's disease
M04.2	Cryopyrin-associated periodic syndromes (CAPS)
M15.0 - M19.93	Osteoarthritis and allied disorders
M31.4	Aortic arch syndrome [Takayasu]
M31.5 - M31.6	Giant cell arteritis
M47.819	Spondylosis without myelopathy or radiculopathy, site unspecified
M54.30 - M54.42	Sciatica

Code	Code Description
M86.30 - M86.9	Chronic multifocal osteomyelitis [recurrent]
M94.1	Relapsing polychondritis
N96	Recurrent pregnancy loss
003.0 - 003.9	Spontaneous abortion
026.20 - 026.23	Pregnancy care for patient with recurrent pregnancy loss

Background

U.S. Food and Drug Administration (FDA)-Approved Indications

Adalimumab

Note: Per labeling, some biosimilar products are not interchangeable or automatically substitutable with Humira. Please see Dosing and Administration section, Appendix, and Full Prescribing Information for more information.

- Rheumatoid Arthritis (RA): reducing signs and symptoms, inducing major clinical response, inhibiting the progression of structural damage, and improving physical function in adult patients with moderately to severely active RA.
- Juvenile Idiopathic Arthritis (JIA): reducing signs and symptoms of moderately to severely active polyarticular JIA in patients 2 years of age and older.
- Psoriatic Arthritis (PsA): reducing signs and symptoms, inhibiting the progression of structural damage, and improving physical function in adult patients with active PsA.
- Ankylosing Spondylitis (AS): reducing signs and symptoms in adult patients with active AS.
- Crohn's Disease (CD): treatment of moderately to severely active Crohn's disease in adults and pediatric patients 6 years of age and older.
- Ulcerative Colitis (UC): treatment of moderately to severely active ulcerative colitis in adults and pediatric patients 5 years of age and older. Limitations of Use: Effectiveness has not been established in patients who have lost response to or were intolerant to TNF blockers.
- Plaque Psoriasis (Ps): treatment of adult patients with moderate to severe chronic plaque psoriasis who are candidates for systemic therapy or phototherapy, and when other systemic therapies are medically less appropriate.
- Hidradenitis Suppurativa (HS): treatment of moderate to severe hidradenitis suppurativa in patients 12 years of age and older.
- Uveitis (UV): treatment of non-infectious intermediate, posterior, and panuveitis in adults and pediatric patients 2 years of age and older.

Compensial Uses

- Behcet's disease
- Non-radiographic axial spondyloarthritis
- Immune checkpoint inhibitor-related toxicity - inflammatory arthritis
- Oligoarticular juvenile idiopathic arthritis
- Pyoderma gangrenosum

Adalimumab is marketed as Humira (Abbott Laboratories, North Chicago, IL), which is a

recombinant human IgG1 monoclonal antibody that acts by inhibiting tumor necrosis factor alpha, an inflammatory protein that, when produced in excess, plays a key role in the inflammatory responses of some autoimmune diseases. Adalimumab binds specifically to human tumor necrosis factor (TNF- α) and blocks its action by modulating responses induced or regulated by TNF- α . TNF- α is a cytokine that plays an important role in various inflammatory processes including: induction of pro-inflammatory cytokines such as interleukins 1 and 6, enhancement of leukocyte migration by increasing endothelial layer permeability and expression of adhesion molecules by endothelial cells and leukocytes, activation of neutrophil and eosinophil functional activity, induction of acute phase reactants and other liver proteins, as well as tissue degrading enzymes produced by synoviocytes and/or chondrocytes.

Adalimumab (Humira) carries black box warnings for serious infection and malignancy. Patients treated with adalimumab are at increased risk of serious infections leading to hospitalization or death, including tuberculosis (TB), bacterial sepsis, invasive fungal infections (such as histoplasmosis), and infections due to other opportunistic pathogens. Opportunistic infections due to bacterial, mycobacterial, invasive fungal, viral, parasitic, or other opportunistic pathogens including aspergillosis, blastomycosis, candidiasis, coccidioidomycosis, histoplasmosis, legionellosis, listeriosis, pneumocystosis and tuberculosis have been reported with TNF blockers. Patients have frequently presented with disseminated rather than localized disease. Lymphoma and other malignancies, some fatal, have been reported in children and adolescent patients treated with TNF blockers including adalimumab. Post-marketing cases of hepatosplenic T-cell lymphoma (HSTCL), a rare type of T-cell lymphoma, have occurred in adolescent and young adults with inflammatory bowel disease treated with TNF blockers including adalimumab (AbbVie, 2024).

Active TB including reactivation of latent TB has been reported in patients taking Humira. Patients with TB have frequently presented with disseminated or extra-pulmonary disease. Anti-TB treatment of patients with latent TB infection reduces the risk of reactivation in patients receiving adalimumab. However, active TB has developed in patients receiving adalimumab whose screening for latent TB infection was negative. The labeling recommends that patients should be evaluated for TB risk factors and be tested for latent TB prior to initiating adalimumab and during therapy. According to the product labeling, when TB skin testing is performed, an induration size of 5 mm or greater should be considered positive, even if the patient was previously vaccinated with Bacille Calmette-Guerin (BCG). Treatment of latent TB should be initiated prior to therapy with adalimumab. The labeling recommends that physicians should monitor patients receiving adalimumab for signs and symptoms of active TB, including patients who tested negative for latent TB.

Per the Centers for Disease Control and Prevention (CDC), the tuberculin (TB) skin test (also called the Mantoux tuberculin skin test [TST]) and the TB blood test (also called interferon-gamma release assays or IGRAs) can be used to test for M. tuberculosis infection. Additional tests are required to confirm TB disease. "A posterior-anterior chest radiograph is used to detect chest abnormalities. Lesions may appear anywhere in the lungs and may differ in size, shape, density, and cavitation. These abnormalities may suggest TB, but cannot be used to definitively diagnose TB. However, a chest radiograph may be used to rule out the possibility of pulmonary TB in a person who has had a positive reaction to a TST or TB blood test and no symptoms of disease."

According to the CDC's Quality ID #176 measure, tuberculosis (TB) screening should be performed within the 12 months prior to initiating a first course of biologic and/or immune response modifier therapy that carries a warning for potential reactivation of latent infections. This applies to adult patients (18 years and older) who are newly prescribed such therapies and have not received similar treatment in the preceding 15 months. The goal is to ensure that TB testing is documented in the medical record before starting treatment to mitigate the risk of reactivating latent TB infection.

Other warnings and precautions include risk of anaphylaxis or serious hypersensitivity reactions, hepatitis B virus reactivation, demyelinating disease, cytopenias, pancytopenia, heart failure and lupus-like syndrome. Use of TNF blockers, including Humira, may increase the risk of reactivation of hepatitis B virus (HBV) in patients who are chronic carriers of this virus. In some instances, HBV reactivation occurring in conjunction with TNF blocker therapy has been fatal. The majority of these reports have occurred in patients concomitantly receiving other medications that suppress the immune system, which may also contribute to HBV reactivation. Use of TNF blocking agents,

including Humira, has also been associated with rare cases of new onset or exacerbation of clinical symptoms and/or radiographic evidence of central nervous system demyelinating disease, including multiple sclerosis (MS) and optic neuritis, and peripheral demyelinating disease, including Guillain-Barré syndrome.

The labeling states that cases of worsening congestive heart failure (CHF) and new onset CHF have been reported with TNF blockers. Humira has not been formally studied in patients with CHF; however, in clinical studies in CHF of another TNF blocker, a higher rate of serious CHF-related adverse reactions was observed. The labeling recommends to exercise caution when using Humira in patients who have heart failure, and to monitor patients with heart failure carefully.

Invasive fungal infections, including histoplasmosis, coccidioidomycosis, candidiasis, aspergillosis, blastomycosis, and pneumocystosis, have been reported in patients taking Humira. Patients with histoplasmosis or other invasive fungal infections may present with disseminated, rather than localized disease. Antigen and antibody testing for histoplasmosis may be negative in some patients with active infection. The labeling states that empiric anti-fungal therapy should be considered in patients at risk for invasive fungal infections who develop severe systemic illness.

The most common adverse reactions (incidence greater than 10%) include infections (e.g. upper respiratory, sinusitis), injection site reactions, headache and rash.

Concurrent use of anakinra (an interleukin-1 antagonist) and another TNF-blocker, was associated with a greater proportion of serious infections and neutropenia and no added benefit compared with the TNF-blocker alone in patients with RA. Therefore, the combination of Humira and anakinra is not recommended.

Adalimumab is intended for use under the guidance and supervision of a physician. A patient may self-inject Humira if a physician determines that it is appropriate, and with medical follow-up, as necessary, after proper training in subcutaneous injection technique.

It is recommended that the prescription be written by or on the advice of a rheumatologist, pediatric rheumatologist, dermatologist or an individual highly familiar with prescribing and monitoring of specialized biologic response modifiers.

In 2016, the FDA approved Humira biosimilar, Amjevita (adalimumab-atto) (Amgen Inc.), which was launched in the United States in January 2023. Amjevita received interchangeable designation in March 2025.

In 2017, the FDA approved Humira biosimilar Cyltezo (adalimumab-adbm) (Boehringer Ingelheim Pharmaceuticals, Inc.). Cyltezo received interchangeable designation on October 15, 2021.

In 2018, the FDA approved Humira biosimilar Hyrimoz (adalimumab-adaz) (Sandoz Inc.).

In 2019, the FDA approved Humira biosimilars Abrilada (adalimumab-afzb) (Pfizer Inc.) and Hadlima (adalimumab-bwwd) (Organon & Co). Abrilada received interchangeable designation on October 5, 2023. Hadlima (Organon & Co.) and unbranded Hadlima (Samsung Bioepis) received interchangeable designation in 2025, marking the fifth interchangeable Humira biosimilar product.

In 2020, the FDA approved Humira biosimilar Hulio (adalimumab-fkjp) (Mylan Specialty L.P.).

In 2021, the FDA approved Humira biosimilar Yusimry (adalimumab-aqvh) (Meitheal Pharmaceuticals, Inc.).

In 2022, the FDA approved Humira biosimilar Idacio (adalimumab-aacf) (Fresenius Kabi USA, LLC).

In 2023, the FDA approved Humira biosimilar Yuflyma (adalimumab-aaty) (Celltrion USA, Inc.).

In 2024, the FDA approved Humira biosimilar Simlandi (adalimumab-ryvk) (Alvotect USA Inc.). Simlandi also received interchangeable designation.

Biosimilar and Interchangeable Drugs

Biosimilar drugs are FDA-approved biological products that have demonstrated that they

are highly similar without clinically meaningful differences in terms of safety and effectiveness from the biologic reference product. Only minor differences in clinically inactive components are allowable in biosimilar drugs. Some biosimilar drugs may have the 'interchangeable' designation. An interchangeable biological product is a biosimilar that meets additional FDA requirements and may be substituted for the reference product at the pharmacy without the intervention of the prescribing health care provider, similar to how generic drugs are routinely substituted for brand-name drugs (FDA, 2023; FDA, undated). For approval as an interchangeable product, the biosimilar must be expected to produce the same clinical result as the reference product in any given patient, and for a product administered more than once, the risk of switching between a reference product and an interchangeable product must not be greater than the risk of using the reference product without switching (Herndon et al., 2023).

Despite the adoption of biosimilars in many therapeutic areas, concerns persist regarding switching a patient to a biosimilar whose condition is stable while on the reference biologic. Herndon et al (2023) performed a systematic review and meta-analysis to examine the occurrence of safety events following a switch to or from a biosimilar and its reference biologic in all identified controlled clinical studies that included a biosimilar approved by the FDA. The authors reviewed FDA databases containing publicly available information for all randomized controlled studies and extension studies with a switch treatment period (STP) to or from a biosimilar and its reference biologic. Information not captured in FDA reviews was identified by searching the Embase, MEDLINE, and PubMed databases for the period of January 1, 2000, through December 31, 2022. Forty-four STPs were identified from 31 unique studies for 21 different biosimilars. Data were extracted and synthesized following PRISMA guidelines. Meta-analysis was conducted to estimate the overall risk difference across studies. A total of 5,252 patients who were switched to or from a biosimilar and its reference biologic were identified. Safety data including deaths, serious adverse events (SAEs), and treatment discontinuation showed an overall risk difference (95% CI) of -0.00 (-0.00, 0.00), 0.00 (-0.01, 0.01), -0.00 (-0.01, 0.00) across STPs, respectively. Immunogenicity data showed similar incidence of anti-drug antibodies and neutralizing antibodies in patients within a STP who were switched to or from a biosimilar to its reference biologic and patients who were not switched. Immune related adverse events such as anaphylaxis, hypersensitivity reactions, and injections site reactions were similar in switched and non-switched patients. The authors report that this is the first systematic review using statistical methods to address the risk of switching patients between reference biologics and biosimilars which found no difference in the safety profiles or immunogenicity rates in patients who were switched and those who remained on a reference biologic or a biosimilar. It is of note that endpoints related to pharmacokinetics (PK) or effectiveness of a drug that were collected during the STPs were not included in this systematic review. If efficacy data were collected periodically during STPs, it was not powered for an efficacy determination. The efficacy data used to determine equivalent efficacy of a biosimilar to a reference biologic is contained in publicly available FDA reviews and in the publications for the individual studies that led to approval of the biosimilars.

In summary, all biological products are approved only after they meet the FDA's rigorous approval standards. This means health care providers and patients can expect the same safety and effectiveness from both a biosimilar and an interchangeable biosimilar, just as they would for a reference product (FDA, 2023; FDA, undated).

The FDA previously recommended that Prescribing Information (labeling) include a "Biosimilarity Statement" or "Interchangeability Statement" describing the product's relationship to its reference product in the Highlights section. However, in 2024, the FDA recommended that all labeling for biosimilars, including interchangeable biosimilars, include one statement - a "biosimilarity statement". The FDA state that statements in the Prescribing Information identifying that products have been approved as interchangeable with the reference product and describing the interchangeability standard are not necessary for informing the safe and effective use of the product to prescribing health care professionals. Moreover, there may be inaccurate perceptions that interchangeable biosimilars are safer or more effective than biosimilars that are not approved as interchangeable. However, the interchangeability designation does not indicate a higher level of biosimilarity. Health care professionals can prescribe both biosimilar and interchangeable biosimilar products in place of the reference product with equal confidence that they are as safe and effective as their reference products. Thus, including biosimilarity statements in the Prescribing Information for both biosimilar and interchangeable biosimilar products may avoid confusion among health care professionals about the significance of FDA's determination that a biosimilar is interchangeable with

the reference product (Yim, 2024).

"FDA recommends that biosimilar and interchangeable biosimilar product labeling incorporate relevant data and information from the reference product labeling, with appropriate modifications. The relevant data and information from the reference product labeling that should be incorporated into the biosimilar and interchangeable biosimilar product labeling will depend on whether the applicant is seeking licensure for all conditions of use (e.g., indication(s), dosing regimen(s)) or fewer than all conditions of use for which the reference product has been previously licensed. In addition, biosimilar and interchangeable biosimilar product labeling may include information specific to the biosimilar or interchangeable biosimilar product that is necessary to inform safe and effective use, including preparation, administration, and storage conditions. This information may differ from that of the reference product labeling to reflect differences between the biosimilar or interchangeable biosimilar product and the reference product" (Yim, 2024).

Biosimilar product labeling may include fewer indications than the reference product if the reference product has exclusivity rights for an indication. This can be due to patent rights or other unexpired exclusivity rights. However, "Prescribers can prescribe both biosimilar and interchangeable biosimilar products in place of the reference product with equal confidence that they are as safe and effective as their reference products" (Yim, 2024).

Alopecia Areata

In a prospective, open-label, single-center, pilot study, Bolduc and Bissonnette (2012) examined the efficacy and safety of adalimumab in patients with severe alopecia areata (AA). Three subjects of the planned 10 were enrolled and received 2 weekly subcutaneous (SC) loading doses of adalimumab 80 mg followed by 40 mg SC every week for 6 months. Patients were evaluated for efficacy and safety on a monthly basis. Enrolment in this trial was stopped following publication of studies showing no improvement in patients with AA treated with tumor necrosis factor α antagonists. One patient had a favorable response to adalimumab, whereas the other 2 patients had no benefit from the therapy. Adalimumab was well-tolerated by patients with AA. The author concluded that adalimumab was well-tolerated in patients with AA; but did not induce clinically significant hair regrowth.

An UpToDate review on "Approach to the patient with a scalp eruption" (Goldstein and Goldstein, 2014) noted that dissecting cellulitis of the scalp presents as deep-seated nodules, pustules, and cysts in adults. These can heal with scarring, sinus tract formation, and significant, often permanent, alopecia. The culture is frequently negative or grows normal skin flora. There is vigorous granulomatous inflammation in this condition, which is typically managed with chronic oral antibiotics or isotretinoin. The review does not mention adalimumab as a therapeutic option.

Gorcey, et al. (2014) stated that alopecia universalis is an uncommon form of AA involving hair loss over the entire scalp and body and is often difficult to treat. Tumor necrosis factor (TNF) inhibitors have been largely unsuccessful in treating AA and have been reported to induce or worsen AA in patients. These investigators reported a case of alopecia universalis successfully treated with adalimumab and discussed the possible mechanism. A woman in her 30s with alopecia universalis, refractory to multiple treatment modalities, was successfully treated with adalimumab. The authors concluded that tumor necrosis factor has multiple important roles in the pathogenesis of AA, and its interplay with other cytokines, specifically interferons, may be responsible for the development of AA in patients treated with TNF inhibitors. Pharmacogenetics and the inherent physiologic levels of TNF may explain why TNF inhibitors cause AA in some individuals, while treating AA in others. They stated that these conclusions warrant further investigation on this subject.

Furthermore, an UpToDate review on "Management of alopecia areata" (Messenger, 2017) states that "Tumor necrosis factor (TNF)-alpha inhibitors have not shown efficacy in the treatment of alopecia areata. An open-label study with etanercept revealed no efficacy of etanercept for the treatment of moderate to severe alopecia areata. In addition, patients who have been treated with adalimumab, infliximab, or etanercept for other autoimmune disorders have developed alopecia areata during the course of treatment."

Ankylosing Spondylitis and Spondyloarthritis

Ankylosing spondylitis (AS) is a form of arthritis known as spondyloarthritis, which is a group of closely linked rheumatic diseases that can cause pain in the spine and joints as well as ligaments and tendons. Ankylosing spondylitis is an autoimmune disorder in which TNF-alpha has been suggested to play a role in its pathogenesis. A chronic disease, AS primarily affects the spine causing back stiffness and potential deformity over time. Wendling and Toussirot (2004) noted that anti-TNF represents a major therapeutic advancement in the treatment of AS. De Keyser and associates (2006) stated that AS is the prototype disease within the spondyloarthropathies, a group of diseases presenting mainly with spondylitis, pauci-articular peripheral arthritis, and enthesiopathy. Non-steroidal anti-inflammatory drugs are the classical cornerstone of medical therapy in these patients; no real DMARD was available, until recently. Tumor necrosis factor-alpha blocking agents (monoclonal antibodies or soluble receptors) are the first representative drugs, of which the indication has recently been expanded to encompass also patients with AS. In a 52-week open-label study (n = 15, mean age of 40 years with a range of 19 to 55 years), Haibel and colleagues (2006) reported that adalimumab treatment of active AS resulted in a clear improvement in clinical (reduction of spinal symptom) as well as MRI outcome measurements, similar to that observed with other TNF-alpha blocking agents.

On July 31, 2006, the FDA granted a supplemental indication for adalimumab to reduce signs and symptoms in patients with active ankylosing spondylitis (AS). This indication was approved by the European Commission in June 2006. The recommended dosage of adalimumab for AS is 40 mg (subcutaneous injection) every other week. The approval of adalimumab for the treatment of patients with active AS is based on data from the ATLAS (Adalimumab Trial Evaluating Long-Term Efficacy and Safety in AS) trial (n = 315), which was a randomized, placebo-controlled, double-blind, phase III study conducted in Europe and the United States of patients with AS who had failed to respond to NSAIDs or DMARDs. Results at 12 weeks showed that 58% of patients receiving adalimumab achieved and sustained a minimum 20% reduction in pain and inflammation, as measured via the Assessment in AS (ASAS) International Working Group criteria for evaluating function, pain, patient global assessment, and inflammation. At week 24, 42% of adalimumab-treated patients versus 16% of those receiving placebo achieved a reduction of 50% or more in disease activity, as evaluated using a patient-assessed composite index for pain, stiffness, and fatigue (Bath AS Disease Activity Index [BASDAI]). Moreover, approximately 1 of 5 patients achieved partial remission, defined as a value of less than 20 on a 0 to 100 scale in each of the 4 ASAS domains.

ATLAS also explored the impact of adalimumab on enthesitis, a primary pathology in AS that is characterized by inflammation of the ligaments attachment to the bone. At week 24, the mean change in the enthesitis symptom score as indexed by Maastricht Ankylosing Spondylitis Enthesitis Score (MASES) in patients treated with adalimumab showed significant reduction. MASES is an index that assesses enthesitis in certain locations, such as the rib cage, lower back, and Achilles tendons.

The British Society for Rheumatology (BSR, 2006) has stated that "there is evidence to support the use of adalimumab as a treatment for adult patients with active AS, who have had an inadequate response to non-steroidal anti-inflammatory drugs and conform to the current BSR guideline for the use of anti-TNF-alpha drugs in AS." The BSR statement notes that "[w]hilst there have not been any direct comparisons between anti-TNF- α drugs in AS, adalimumab appears to be as effective as any other licensed agents."

According to the Spanish Society of Rheumatology's Clinical Guidelines for Patients with Spondyloarthritis (2010), spondyloarthritis (SpA) includes: ankylosing spondylitis (AS); reactive arthritis (ReA); arthritis-related inflammatory bowel diseases (IBD), Crohn's disease and ulcerative colitis; psoriatic arthritis (PsA); undifferentiated SpA (uSpA); and juvenile-onset AS (JoAS). The guideline stated that the use of adalimumab (ADA) is recommended in axial AS patients refractory to NSAIDs and other symptomatic treatments (analgesics, local GC infiltrations), and in peripheral AS patients refractory to NSAIDs, DMARDs, and symptomatic treatments (analgesics, local GC infiltrations, synoviorthesis).

UpToDate defines spondyloarthritis as a group of arthritis-associated conditions. One of the associated conditions includes axial spondyloarthritis (axSpA), which is a potentially disabling inflammatory arthritis that typically affects the spine; however, it may also affect joints in the arms and legs. Symptoms typically begin before the age of 45. Those with axSpA can be further classified into 2 subtypes such as ankylosing spondylitis (AS) and nonradiographic axial spondyloarthritis (nr-axSpA). Individuals affected by AS exhibit radiographic abnormalities consistent with sacroiliitis, whereas findings are absent or minimal on plain radiography for those with nr-axSpA. Individuals with nr-axSpA may have evidence of active inflammation of the sacroiliac (SI) joints on

MRI findings (e.g., bone marrow edema of the joints). The diagnosis of axSpA, including AS and nr-axSpA, should be considered in persons with continuous chronic back pain prior to age 45. Per UpToDate, "there is no single historical feature, physical finding, laboratory test, or imaging study with sufficient specificity by itself to establish the diagnosis without the presence of additional abnormalities. Thus, the presence of a combination of features together with the exclusion of other diagnoses that may explain such symptoms or findings is necessary to arrive at an accurate diagnosis" (Yu and van Tubergen, 2018a).

Sieper, et al. (2013) evaluated the safety and effectiveness of adalimumab in patients with non-radiographic axial spondyloarthritis (nr-axSpA). Patients fulfilled Assessment of Spondyloarthritis International Society (ASAS) criteria for axial spondyloarthritis, had a Bath Ankylosing Spondylitis Disease Activity Index (BASDAI) score of greater than or equal to 4, total back pain score of greater than or equal to 4 (10 cm visual analog scale), and inadequate response, intolerance, or contraindication to non-steroidal anti-inflammatory drugs (NSAIDs); patients fulfilling modified New York criteria for ankylosing spondylitis were excluded. Patients were randomized to adalimumab (n = 91) or placebo (n = 94). The primary endpoint was the percentage of patients achieving ASAS40 at week 12. Efficacy assessments included BASDAI and Ankylosing Spondylitis Disease Activity Score (ASDAS). MRI was performed at baseline and week 12 and scored using the Spondyloarthritis Research Consortium of Canada (SPARCC) index. Significantly more patients in the adalimumab group achieved ASAS40 at week 12 compared with patients in the placebo group (36% versus 15%, $p < 0.001$). Significant clinical improvements based on other ASAS responses, ASDAS, and BASDAI were also detected at week 12 with adalimumab treatment, as were improvements in quality of life measures. Inflammation in the spine and sacroiliac joints on MRI significantly decreased after 12 weeks of adalimumab treatment. Shorter disease duration, younger age, elevated baseline C-reactive protein, or higher SPARCC MRI sacroiliac joint scores were associated with better week 12 responses to adalimumab. The safety profile was consistent with what is known for adalimumab in ankylosing spondylitis and other diseases. The authors concluded that in patients with nr-axSpA, adalimumab treatment resulted in effective control of disease activity, decreased inflammation, and improved quality of life compared with placebo. They stated that results from ABILITY-1 suggested that adalimumab has a positive benefit-risk profile in active nr-axSpA patients with inadequate response to NSAIDs.

Paramarta, et al. (2013) evaluated the safety and effectiveness of adalimumab in patients with peripheral SpA not fulfilling the criteria for ankylosing spondylitis (AS) or psoriatic arthritis (PsA). A total of 40 patients with active peripheral SpA fulfilling the European Spondyloarthropathy Study Group or Amor criteria but not the criteria for AS or PsA were included in a randomized, double-blind, placebo-controlled clinical trial. Patients were treated 1 to 1 with adalimumab or placebo for 12 weeks, followed by an open-label extension up to week 24. Safety and efficacy measurements were performed every 6 weeks, with the patient's global assessment of disease activity at week 12 as the primary endpoint. At week 12, the patient's and physician's global assessment of disease activity, swollen joint count, BASDAI, ASDAS, and erythrocyte sedimentation rate improved significantly in the adalimumab group compared with the baseline values and compared with placebo. A similar improvement was seen upon adalimumab treatment from weeks 12 to 24 in the patients originally randomized to placebo, whereas the clinical response was maintained or even augmented at week 24 in the patients who received adalimumab from the start. ASDAS inactive disease and BASDAI50 responses were met in 42% of the adalimumab group versus 0% to 5% in the placebo group at week 12 ($p = 0.001$ and $p = 0.008$, respectively), and were further increased at week 24. The number of adverse events was not different between the adalimumab and placebo groups. The authors concluded that adalimumab appeared to be effective and well-tolerated in SpA patients with peripheral arthritis, also in those patients not fulfilling the AS or PsA criteria.

Per the Prescribing Information of Humira, adalimumab is indicated for reducing signs and symptoms in adult patients with active ankylosing spondylitis. Although spondyloarthritis is not listed as an indication of Humira (AbbVie, 2020), UpToDate recommends adding a tumor necrosis factor (TNF)-alpha inhibitor for patients with active axial SpA symptoms and have had an inadequate response to initial therapy with at least two NSAIDs consecutively rather than treatment with NSAIDs alone. TNF inhibitors (e.g., subcutaneous adalimumab) is an acceptable treatment option (Yu and van Tubergen, 2019b).

Chronic Recurrent Multifocal Osteomyelitis (CRMO)

The Arthritis Research UK's "The use of pamidronate and adalimumab in children with

chronic recurrent multifocal osteomyelitis" (2014) concluded that "Overall, this report identifies a large body of case studies regarding the use of pamidronate and/or adalimumab in the treatment of pediatric or adult chronic recurrent multifocal osteomyelitis (CNO, SAPHO, CRMO), however, whilst two intervention studies were identified these were both uncontrolled. These findings concur with those reported in the systematic reviews and consensus statements included in this evidence review. In general, it appears that pamidronate and adalimumab are beneficial, at least in some cases; however, there is a paucity of well-designed controlled trials to substantiate these findings and future trials are therefore warranted".

Kaiser and colleagues (2015) determined the clinical presentation, current treatment and outcome of children with non-bacterial inflammatory bone disease. Of the 8 patients treated with biologics, only 2 were treated successfully for osteomyelitis with etanercept over a period of 12 and 28 months, respectively. One patient had adalimumab for arthritis but had persistent osteomyelitis despite improvement of arthritis. In 2 cases treatment with TNF-blockers was stopped after a short time because of side effects (skin infection, allergic reaction) and in further 2 cases treatment was unsuccessful. One patient had anakinra without improvement.

Crohn's Disease

Adalimumab has also been shown in clinical trials to be effective for moderate to severe Crohn's disease. Colombel, et al. (2007) reported on the results of a controlled clinical trial, which demonstrated that adalimumab was effective in maintenance of response and remission in patients with moderate to severe Crohn's disease. In this clinical study, patients received open-label induction therapy with adalimumab 80 mg at study initiation followed by 40 mg 2 weeks later. Four weeks following study initiation, patients who responded to adalimumab were stratified by response (decrease in Crohn's Disease Activity Index greater than or equal to 70 points from baseline) and randomized to 3 treatment groups:

Patients were followed for 56 weeks. Co-primary end points were the percentages of randomized responders who achieved clinical remission (Crohn's Disease Activity Index score less than 150) at weeks 26 and 56. The investigators reported that the percentage of randomized responders in remission was significantly greater in the groups receiving adalimumab weekly and every other week compared to the placebo group at week 26 (47 %, 40 %, and 17 %, respectively; $p < 0.001$) and at week 56 (41 %, 36 %, and 12 %, respectively; $p < 0.001$). There were no significant differences in efficacy between adalimumab administered weekly and every other week. The investigators noted that adalimumab was well-tolerated; more patients receiving placebo discontinued treatment because of an adverse event (13.4 %) than those receiving adalimumab (4.7 % and 6.9 % in the adalimumab weekly and every other week groups, respectively). The investigators concluded that, among patients who responded to adalimumab, both weekly and every other week adalimumab was significantly more effective than placebo in maintaining remission in moderate to severe Crohn's disease through 56 weeks.

Hanauer, et al. (2006) reported that adalimumab was shown in a controlled clinical trial to be superior to placebo for induction of remission in patients with moderate to severe Crohn's disease. A total of 299 patients with moderate to severe Crohn's disease naive to anti-TNF therapy were randomized to receive subcutaneous injections at study initiation and 2 weeks later with adalimumab 40 mg/20 mg, 80 mg/40 mg, or 160 mg/80 mg or placebo. The primary endpoint was demonstration of a significant difference in the rates of remission (defined as a Crohn's Disease Activity Index score less than 150 points) at 4 weeks after study initiation among the 80 mg/40 mg, 160 mg/80 mg, and placebo groups. The investigators found that the rates of remission at the 4th week in the adalimumab 40 mg/20 mg, 80 mg/40 mg, and 160 mg/80 mg groups were 18 % ($p = 0.36$), 24 % ($p = 0.06$), and 36 % ($p = 0.001$), respectively, and 12 % in the placebo group. The investigators reported that adalimumab was well-tolerated, noting that adverse events occurred at similar frequencies in all 4 treatment groups except injection site reactions, which were more common in adalimumab-treated patients. The investigators concluded that adalimumab was superior to placebo for induction of remission in patients with moderate to severe Crohn's disease naive to anti-TNF therapy. The investigators stated that the optimal induction dosing regimen for adalimumab in this study was 160 mg at initiation of therapy followed by 80 mg 2 weeks later.

Colombel, et al. (2009) compared outcomes of induction dosing followed by continuous

adalimumab treatment with those of induction dosing with re-initiation of adalimumab (in the event of clinical deterioration) for patients with moderate-to-severe Crohn's disease who participated in the Crohn's Trial of the Fully Human Antibody Adalimumab for Remission Maintenance (CHARM). In the CHARM trial, all patients received open-label induction therapy with adalimumab 80 mg and 40 mg at weeks 0 and 2, respectively. In total, 778 patients were randomized at week 4 to one of three groups:

At/after week 12, patients receiving placebo with flare or non-response could re-initiate open-label adalimumab 40 mg e.o.w., and patients receiving continuous blinded adalimumab therapy could switch to open-label 40 mg e.o.w. Patients in all groups could switch to weekly therapy with continued flare/non-response. In the previously published primary analysis, results for only those patients who had responded at week 4 (decrease in Crohn's Disease Activity Index (CDAI) of greater than or equal to 70 points, referred to as "randomized responders") and remained on blinded therapy were analyzed. In this analysis, data from all randomized patients were analyzed based on original randomized treatment using an intention-to-treat analysis, regardless of whether they subsequently switched to open-label therapy. Disease activity, clinical remission, number of flares, Inflammatory Bowel Disease Questionnaire (IBDQ) score, number of Crohn's disease-related surgeries, and hospitalization incidence were compared between the continuous and induction only/reinitiation adalimumab groups. Results for all outcome measures were superior for both continuous groups compared with the induction only/re-initiation group. On the basis of median CDAI and IBDQ results, patients in both continuous treatment groups achieved statistically significantly greater improvements versus the induction only/re-initiation group ($p < 0.05$). At week 56, a significantly greater percentage of patients who had received continuous adalimumab (51 % for e.o.w. and 49 % for weekly) were in clinical remission versus the induction only/re-initiation group (38 %, $p < 0.05$). Continuous adalimumab therapy was also associated with fewer flares and fewer Crohn's disease-related surgeries ($p < 0.05$). Patients in both continuous adalimumab groups had significantly lower risks of Crohn's disease-related and all-cause hospitalizations than did patients in the induction only/re-initiation group ($p < 0.05$). The authors concluded that for patients with active Crohn's disease, continuous treatment with adalimumab was more effective than a strategy of induction dosing followed by re-initiation of adalimumab with clinical deterioration for maintenance of clinical remission, improved quality-of-life outcomes, reduced flares, and a decrease in number of surgeries and risk of hospitalization.

Available evidence indicates that adalimumab, infliximab, and certolizumab are comparably effective for Crohn's disease. In a meta-analysis, Kawalec and colleagues (2013) compared the effectiveness and safety of tumor necrosis factor- α (TNF- α) antibodies (infliximab, adalimumab, and certolizumab) with either a placebo or each of them in the treatment of Crohn's disease (CD). A systematic review of literature published up to November 2012 was performed, and a meta-analysis of identified studies was carried out. These investigators searched the following databases: PubMed, EMBASE, The Cochrane Library, and others. Only randomized or clinical controlled trials were included. A total of 19 clinical trials fulfilled the established criteria (5 studies for infliximab versus placebo, 6 for each adalimumab or certolizumab versus placebo, and 2 comparing infliximab with adalimumab). The results of the meta-analysis showed that anti-TNF therapy in patients with CD is safe and statistically significantly more effective when compared with the placebo for induction of remission at week 4 (RB = 1.90, 95% confidence interval [CI]: 1.55 to 2.33, $p < 0.00001$), maintenance of remission at weeks 20 to 30 (RB = 1.86, 95% CI: 1.61 to 2.15, $p < 0.00001$), and at weeks 48 to 56 (RB = 2.75, 95% CI: 2.13 to 3.54, $p < 0.00001$) in patients who responded to the induction therapy and patients randomized before the induction. Anti-TNF agents were also superior to the placebo in fistula healing (during short-term induction, as well as long-term maintenance) and inducing CR-70 but not CR-100 at week 4. Moreover, the anti-TNF therapy had a significant effect on achieving both CR-70 and CR-100 during long-term maintenance. The authors concluded that infliximab, adalimumab, and certolizumab are effective as both induction and maintenance therapy in moderate to severe Crohn's disease in adults, including patients with fistulas. The safety profile was acceptable.

Patil and associates (2013) compared the efficacy of the 3 anti-TNFs for CD in clinical practice. Retrospective review of patients initiated on anti-TNF between 2004 and 2008 was carried out. Disease activity, quality of life, and remission rates were compared between groups over 1 year. A total of 60 patients with CD were initiated on anti-TNF from 2004 to 2008: 31 on infliximab (IFX) and 29 on adalimumab (ADA) or certolizumab pegol (CTZ). More patients in the ADA/CTZ scores group had prior exposure to anti-TNF

(76% versus 10%, $p < 0.01$). Mean Harvey-Bradshaw Index (HBI) scores in the IFX group were lower than in the ADA/CTZ group at 12 months (2.72 ± 3.34 versus 5.63 ± 5.33 , $p = 0.03$). At 12 months, more IFX patients were in remission compared with those on ADA/CTZ (88% versus 53%, $p \leq 0.01$). Mean short IBD questionnaire (SIBDQ) scores were not different between the IFX and ADA/CTZ groups at 12 months. Stratified analyses and logistic regression based on prior anti-TNF use did not show differences in remission rates at any time point post-baseline between groups. The authors concluded that after adjustment for prior anti-TNF, there was no difference in remission rates between the IFX and ADA/CTZ groups at any time point post-baseline. This suggested that differences between groups were accounted for by a higher rate of prior anti-TNF in the ADA/CTZ group. The authors noted that these results should be reviewed with caution given the small sample size.

Mozaffari, et al. (2013) noted that some inflammatory bowel disease (IBD) patients, especially those with refractory CD or relapsing ulcerative colitis (UC), do not respond to current therapies. The newly introduced biological drugs have garnered interest due to their specificity and selectivity in modulation of inflammatory elements. In 46 randomized, placebo-controlled clinical trials, the efficacy and safety of different biologic drugs were evaluated in moderately to severely active CD or UC patients. Current investigated drugs include new anti-TNF drugs (adalimumab, certolizumab pegol, etanercept, onercept, and golimumab), anti-CD20 (rituximab), T-cell inhibitors (abatacept), and anti- $\alpha 4$ integrins (natalizumab and vedolizumab). Adalimumab, certolizumab, and golimumab showed significant efficacy in induction of remission and maintenance in CD and UC patients with a rate of adverse events similar to placebo in the major trials. Natalizumab and vedolizumab were effective in the treatment of moderately to severely active CD and UC patients. However, vedolizumab caused fewer adverse effects than natalizumab. Onercept, etanercept, rituximab, and abatacept were all well-tolerated but were not effective in CD or UC patients. The authors concluded that anti-TNF drugs, except for onercept and etanercept, and anti- $\alpha 4$ integrins exhibit beneficial therapeutic effects. Although they were all well-tolerated, the incidence of progressive multifocal leukoencephalopathy associated with natalizumab should not be overlooked.

Krznaric, et al. (2013) stated that biological therapy (infliximab and adalimumab) in IBD is based on the IgG1 anti-TNF monoclonal antibodies with potent anti-inflammatory effects, whose main mechanism of action is thought to be the induction of inflammatory cell apoptosis. Unquestionably, which arises from the most recent studies and meta-analysis, anti-TNF agents are an effective therapy primarily for the treatment of CD, but also UC, in different clinical situations. Infliximab has the most extensive clinical trial data, but other biological agents, such as adalimumab and certolizumab pegol, appear to have similar benefits. In terms of future research, more long-term data are needed for both certolizumab pegol in CD and adalimumab in UC. An important role in the application of biological therapy is assessing its effectiveness and cost-benefit relationships that are estimated by regular follow-up. In the absence of response (primary and secondary), therapeutic options are dose increase, giving the drug in shorter intervals, and substitution with another biological drug.

Adalimumab has been shown to be effective for extraintestinal manifestations of Crohn's disease. Barrie and Regueiro (2007) stated that inflammatory bowel diseases (IBD), notably Crohn's disease (CD) and ulcerative colitis (UC), are systemic inflammatory diseases primarily involving the gastrointestinal tract. Twenty percent to 40% of patients with IBD develop extra-intestinal inflammation and symptoms, known as extra-intestinal manifestations (EIMs). The most common EIMs affect the joints, skin, eyes, and biliary tract. The EIMs associated with IBD bear a negative impact on patients with UC and CD. Thus, the successful treatment of EIMs is essential for improving the quality of life of IBD patients. For most EIMs, their resolution often parallels that of the active IBD in both timing and therapy required. However, some EIMs, such as axial arthritis, pyoderma gangrenosum, uveitis, and primary sclerosing cholangitis, run a clinical course independent of IBD disease activity. The advent of biologic response modifiers, e.g., tumor necrosis factor- α (TNF) inhibitors, has improved the treatment of IBD and its associated EIMs. These investigators reviewed the therapeutic experiences of the 2 most widely used anti-TNF neutralizing antibodies, infliximab and adalimumab, for immune-mediated EIM of IBD.

Zippi, et al. (2013) stated that UC and CD are the 2 forms of IBD. The advent of biological drugs has significantly changed the management of these conditions. Skin manifestations are not uncommon in IBD. Among the reactive lesions (immune-mediated EIMs), erythema nodosum (EN) and pyoderma gangrenosum (PG) are the 2 major cutaneous ills associated with IBD, while psoriasis is the dermatological comorbidity disease observed

more often. In particular, in the last few years, anti-TNF- α agents have been successfully used to treat psoriasis, especially these kinds of lesions that may occur during the treatment with biological therapies. The entity of the paradoxical manifestations has been relatively underreported as most lesions are limited and a causal relationship with the treatment is often poorly understood. The reason for this apparent side effect of the therapy still remains unclear. Although side effects may occur, their clinical benefits are undoubted. The authors reviewed the therapeutic effects of the 2 most widely used anti-TNF- α molecules, infliximab (a fusion protein dimer of the human TNF- α receptor) and adalimumab (a fully human monoclonal antibody to TNF- α), for the treatment of the major cutaneous manifestations associated with IBD (EN, PG, and psoriasis).

Vavricka, et al. (2014) noted that EIM in IBD occur frequently and may present before or after IBD diagnosis. They most commonly affect the eyes, skin, and joints, but can also involve other organs such as the liver. Some EIMs are associated with intestinal disease activity and ameliorate by treatment of the underlying IBD. This is seen in patients with peripheral type 1 arthritis, oral aphthous ulcers, episcleritis, and erythema nodosum. Other EIMs are intestinal disease activity-independent, such as uveitis and ankylosing spondylitis. Finally, some EIMs (e.g., pyoderma gangrenosum and primary sclerosing cholangitis) may or may not be associated with the underlying IBD. Successful therapy of EIMs is important for improving the quality of life of IBD patients. TNF antibody therapy is an important treatment option for EIM in IBD patients, whereas no such beneficial effect was reported for alpha 4 beta 7 integrin antibodies such as vedolizumab so far. This article reviewed the therapeutic experience with TNF antibodies for the treatment of EIM in IBD patients.

The "ACG Clinical Guideline: Management of Crohn's Disease in Adults" by Lichtenstein, et al. (2025) emphasizes the importance of effective treatment for Crohn's Disease (CD), as only 20%-30% of patients experience a nonprogressive course. It highlights that symptoms alone are not reliable indicators of active inflammation, necessitating regular objective evaluations through endoscopy or imaging. The guideline strongly recommends anti-TNF agents, including adalimumab, for both induction and maintenance of remission in patients with moderately to severely active CD, noting that these therapies can show rapid effects within two weeks. Early intervention with anti-TNF agents is linked to better outcomes, especially if started within two years of diagnosis. Additionally, adalimumab has proven effective in maintaining remission after successful induction, with clinical trials supporting its long-term use. The article underscores the significance of monitoring and assessing treatment efficacy to optimize patient outcomes, particularly for those with multiple risk factors for disease progression.

Cryopyrin-Associated Periodic Syndrome (CAPS) and Inflammatory Polyarthropathy

An UpToDate review on "Cryopyrin-associated periodic syndromes and related disorders" (Nigrovic, 2020) does not mention adalimumab as a therapeutic option.

Cystoid Macular Degeneration

In a prospective, non-comparative, non-randomized, pilot study, Diaz-Llopis and colleagues (2008) evaluated the safety and effectiveness of adalimumab in treating refractory autoimmune uveitis. A total of 19 patients meeting eligibility criteria received a 40-mg subcutaneous (s.c.) injection of adalimumab every other week during 1 year. All patients underwent an outcome assessment at month 12. Visual acuity (VA) improved by -0.3 logMar in 12 (31%) eyes of 38, and worsened by +0.3 logMar in 1 (2.6%) eye. All patients had an active intra-ocular inflammation at baseline, and 12 patients (63%) achieved control of their inflammation with adalimumab at the end of follow-up. After optic coherence tomography (OCT), 33 eyes (86%) had cystoid macular edema (CME) at baseline, and at the end of follow-up, there was a complete resolution of CME in 18 of these 33 eyes (54.54%). All patients were able to reduce at least 50% of the dose of the concomitant immunosuppressive drugs at the end of follow-up. Adalimumab was well-tolerated in all patients, and only local minor side effects at the s.c. injection site were observed. Nevertheless, 8 patients (42.10%) had relapses during the follow-up period that were controlled with 1 peri-ocular steroid injection. The authors concluded that adalimumab appeared to be a safe and effective therapy for the management of refractory uveitis and may provide the possibility to reduce the concomitant immunosuppressive drugs in these patients. They stated that further long-term studies are needed to determine the safety and effectiveness of adalimumab in treating intra-ocular inflammation.

Androudi, et al. (2010) evaluated the safety and effectiveness of intra-vitreous adalimumab injections on refractory cystoid macular edema (CME) secondary to non-infectious uveitis. A total of 8 consecutive patients with controlled uveitis and chronic, refractory CME who had failed steroid treatment were included in this study. Intra-vitreous adalimumab injections were given monthly for 3 months. The main outcome measure was the mean change in central retinal thickness (CRT) on optical coherence tomography (OCT); the secondary objective was the mean change in best-corrected visual acuity (BCVA). Five of the 8 patients completed the 6-month follow-up. For all 5 patients, the changes in BCVA from baseline to 3 months were not statistically significant ($p = 0.070$). Similarly, the change in BCVA from baseline to 6 months was not statistically significant ($p = 1.0$). The mean CRT at baseline was 692 microm. The changes from baseline to 3 months were not statistically significant ($p = 0.466$); the changes from baseline to 6 months were also not statistically significant ($p = 0.808$). These researchers did not observe any ocular or systemic adverse effects. The authors concluded that intra-vitreous adalimumab showed no efficacy in improving BCVA or reducing CRT in patients with chronic uveitic macular edema. Furthermore, Neri et al. (2011) stated that adalimumab is a promising drug for the treatment of uveitis, although further studies are needed on its application in uveitis.

In a prospective, non-comparative, interventional case-series study, Androudi, et al. (2010) evaluated the safety and effectiveness of intra-vitreous adalimumab injections on refractory CME secondary to non-infectious uveitis. A total of 8 consecutive patients with controlled uveitis and chronic, refractory CME who had failed steroid treatment were included in this study. Intra-vitreous adalimumab injections were given monthly for 3 months. The main outcome measures were mean change in central retinal thickness (CRT) on OCT; the secondary objective was the mean change in best-corrected VA (BCVA). Five of the 8 patients completed the 6-month follow-up. For all 5 patients, the changes in BCVA from baseline to 3 months were not statistically significant ($p = 0.070$). Similarly, the change in BCVA from baseline to 6 months was not statistically significant ($p = 1.0$). The mean CRT at baseline was 692 microm. The changes from baseline to 3 months were not statistically significant ($p = 0.466$); the changes from baseline to 6 months were also not statistically significant ($p = 0.808$). These researchers did not observe any ocular or systemic adverse effects. The authors concluded that intra-vitreous adalimumab showed no efficacy in improving BCVA or reducing CRT in patients with chronic uveitic macular edema.

Schaap-Fogler, et al. (2014) evaluated the effectiveness of anti-tumor necrosis factor- α (TNF- α) agents for the treatment of uveitis-related refractory CME. The files of 23 consecutive patients treated for CME at the uveitis services of 2 tertiary medical centers from 2006 to 2011 were reviewed for demographic data, VA, and OCT measurements at baseline and 3, 6, and 12 months after treatment. Changes in mean VA and macular thickness were analyzed. Findings were compared between patients treated with a conventional immunosuppressive regimen only ($n = 18$, 27 eyes) and patients treated with an anti-TNF- α agent after proving refractory to conventional treatment ($n = 9$, 15 eyes). Mean duration of CME before initiation of anti-TNF- α therapy was 12 ± 8 months. The 2 groups had similar baseline values of mean central macular thickness (CMT) and VA. Significant improvement in macular thickness was noted at 3 months in both groups, with a maximal effect at 6 months for the anti-TNF- α group ($p = 0.002$). Maximal improvement in VA was achieved at 3 months in both groups, with a reduced effect towards 12 months. Mean VA at 12 months was similar in both groups. The authors concluded that anti-TNF- α agents may serve as an effective lasting treatment for long-standing refractory uveitis-related CME. Moreover, they stated that the role of anti-TNF- α agents as first-line therapy in this setting warrants further investigation. This was a small ($n = 9$) study, and it is unclear (from the abstract) how many patients received adalimumab.

Furthermore, an UpToDate review on "Age-related macular degeneration: Treatment and prevention" (Arroyo, 2017) does not mention adalimumab as a therapeutic option.

Giant Cell Arteritis

In a double-blind, multi-center RCT, Seror, et al. (2014) evaluated the effect of adding a 10-week treatment of adalimumab to a standardized treatment with corticosteroids on the ability to taper more rapidly corticosteroid doses in patients with newly diagnosed giant cell arteritis (GCA). Patients were randomly assigned to receive a 10-week subcutaneous treatment of adalimumab 40 mg every other week or placebo in addition to a standard prednisone regimen (starting dose 0.7 mg/kg per day). The primary end-point was the percentage of patients in remission on less than 0.1 mg/kg of prednisone at week 26.

Analysis was performed by intention-to-treat (ITT). Among the 70 patients enrolled (adalimumab, n = 34; placebo, n = 36), 10 patients did not receive the scheduled treatment, 7 in the adalimumab and 3 in the placebo group. By ITT, the number of patients achieving the primary end-point was 20 (58.9 %) and 18 (50.0 %) in the adalimumab and placebo arm, respectively (p = 0.46). The decrease in prednisone dose and the proportion of patients who were relapse-free did not differ between the 2 groups. Serious adverse events occurred in 5 (14.7 %) patients on adalimumab and 17 (47.2 %) on placebo, including serious infections in 3 patients on adalimumab and 5 on placebo. Two patients died in the placebo arm (septic shock and cancer) and 1 in the adalimumab group (pneumonia). The authors concluded that in patients with newly diagnosed GCA, adding a 10-week treatment of adalimumab to prednisone did not increase the number of patients in remission on less than 0.1 mg/kg of corticosteroids at 6 months.

Granuloma Annulare (Necrobiotic Papulosis)

Mahmood and colleagues (2015) reported the case of a 64-year old woman who demonstrated resolution of recalcitrant, generalized granuloma annulare (GGA) following treatment with adalimumab. After showing little response to other treatment techniques, such as steroids and a triple antibiotic regimen, the patient was started on adalimumab. Within 3 months, she showed almost complete resolution of lesions. Within 6 months, she was completely lesion-free, and remained clear following 12 months of adalimumab therapy. The authors concluded that this case provided further evidence that TNF- α inhibitors may be a therapeutic option for patients with recalcitrant GGA.

Lukacs and associates (2015) stated that GA is a benign inflammatory skin disease. Localized GA is likely to resolve spontaneously, while GGA is rare and may persist for decades. GGA usually is resistant to a variety of therapeutic modalities and takes a chronic course. These investigators summarized all reported treatments of GGA. This is a systematic review based on Medline, Embase and Cochrane Central Register search of articles in English and German and a manual search, between 1980 and 2013, to summarize the treatment of GGA. Most medical literature on treatment of GGA is limited to individual case reports and small series of patients treated without a control group; RCTs are missing. Multiple treatment modalities for GGA were reported including topical and systemic steroids, PUVA, isotretinoin, dapsone, pentoxifylline, hydroxychloroquine, cyclosporine, interferon-gamma (IFN- γ), potassium iodide, nicotinamide, niacinamide, salicylic acid, dipyridamole, PDT, fumaric acid ester, etanercept, infliximab, adalimumab. The authors concluded that while there are numerous case reports of successful treatments in the literature including surgical, medical and phototherapy options, well-designed RCTs are needed for an evidence-based treatment of GGA.

In a single-center, observational study, Min and Lebwohl (2016) evaluated adalimumab's efficacy in treating GGA or disseminated GA (DGA). These researchers identified 7 patients (aged 51 to 77 years) with physician-verified GA who were treated with adalimumab. Primary end-points were improvements in GA Investigator Global Assessment score and BSA; secondary end-points included erythema and induration improvement. All patients' GA Investigator Global Assessment scores improved by 2 or greater. These investigators found significant improvements in BSA, erythema, and induration (average improvements by 87 %, 88 %, and 95 %, respectively); 2 patients required increases in adalimumab frequency; 2 others noted GA recurrence; 1 re-started adalimumab and cleared again. Adverse events (AEs) were minimal. The authors concluded that these findings suggested that adalimumab could be an effective and well-tolerated treatment for GA. It is a particularly attractive therapy for patients who have GGA or DGA. The drawbacks of this study were its small sample size (n = 7) and observational nature of the study.

Chen and co-workers (2019) noted that GA is considered a Th1-type delayed hypersensitivity reaction. Localized GA is likely to resolve spontaneously, whereas DGA may persist for decades and can be resistant to treatment. Biologics including TNF- α inhibitors have been proposed and utilized as salvage therapy for GA and other related diseases, interstitial granulomatous dermatitis (IGD), and actinic granuloma (AG). These investigators carried out a systematic review using the combination of search terms "granuloma annulare", "interstitial granulomatous dermatitis" or "actinic granuloma" and either "biologics", "etanercept", "adalimumab", "infliximab", "ustekinumab", "ixekizumab", "secukinumab", "guselkumab", "golimumab", "brodalumab", "tildrakizumab" or "certolizumab" from the years 1970 to 2017. Review of the literature showed that 79.3 % of the patients with GA, IGD, or AG who had been treated with TNF- α inhibitor therapy showed a clinical response. The authors concluded that TNF- α inhibitor therapy has been used to treat chronic GA, IGD, and AG that involved extensive BSA. However, the

literature was limited to case-series studies that lacked control groups. These researchers stated that RCTs are needed to establish evidence-based treatment of GA and related cutaneous, granulomatous conditions.

Furthermore, an UpToDate review on "Granuloma annulare" (Brodell, 2020) states that "For patients with generalized GA who desire treatment, we suggest topical calcineurin inhibitors, UVA1, or psoralen plus ultraviolet A (PUVA) as first-line treatment (Grade 2C). Topical corticosteroids can be useful for treating select lesions in generalized GA. Limited data suggest that systemic treatment with hydroxychloroquine, isotretinoin, dapsone, adalimumab, and infliximab may be beneficial for generalized GA".

Hidradenitis Suppurativa

U.S. Food and Drug Administration (FDA) approved adalimumab (Humira) for the treatment of moderate to severe hidradenitis suppurativa (HS). This FDA approval is based on the results of two pivotal Phase 3 studies, PIONEER I and PIONEER II, which included 633 people with moderate to severe HS. Patients in these studies were randomly assigned to receive either adalimumab or placebo in addition to daily use of topical antiseptic. Both studies showed that more patients given adalimumab had reductions in the total number of abscesses and inflammatory nodules than patients given placebo. No new safety risks were identified in these trials.

Sotiriou et al. (2012) evaluated the safety and effectiveness of adalimumab using a higher dosage regimen for the treatment of hidradenitis suppurativa (HS) and established the recurrence-free interval after treatment discontinuation. Patients with moderate-to-severe HS were treated with 80 mg adalimumab at baseline, followed by 40 mg every week for 24 weeks. Subsequently, patients entered an observational period for another 24 weeks. Clinical evaluation took place every 4 weeks during the study period. Sartorius scoring system was used as assessment tool regarding disease activity. At the same time points patients evaluated disease activity by visual analog scale (VAS). They completed a Dermatology Life Quality Index (DLQI) questionnaire at baseline and at weeks 24 and 48. A total of 15 patients completed the study. Significant reduction in Sartorius score was obtained by week 24 with a marked improvement during the first month. Mean time to relapse was 11 weeks after treatment cessation, but even at the final visit Sartorius score was significantly lower than at baseline. VAS score and DLQI showed a significant decrease at week 24. There was significant worsening at week 48, however both scores remained significantly lower than baseline levels. The authors concluded that these findings demonstrated the significant efficacy of the once weekly regimen, as well as its benefit regarding time to recurrence. However, they stated that the question if benefit outweighs the risk of a long-term anti-TNF- α antagonist's administration needs still to be answered.

In a phase II, parallel, randomized, placebo-controlled trial consisting of a blinded 16-week period (period 1) and an open-label 36-week period (period 2), Kimball et al (2012) evaluated the safety and effectiveness of adalimumab in patients with moderate to severe HS. A total of 154 adult patients with moderate to severe HS who were unresponsive or intolerant to oral antibiotics were included in this study. Patients were assigned in a 1:1:1 ratio to adalimumab, 40 mg/wk; adalimumab, 40 mg every other week (EOW); or placebo. All patients received adalimumab, 40 mg EOW, at the beginning of period 2 but switched to weekly dosing if the response was suboptimal (HS Physician's Global Assessment [PGA] score of moderate or worse) at weeks 28 or 31. The primary outcome measure (clinical response) was the proportion of patients achieving an HS-PGA score of clear, minimal, or mild with at least a 2-grade improvement relative to baseline at week 16. At week 16, 3.9 % of placebo patients (2 of 51), 9.6 % of EOW patients (5 of 52), and 17.6 % of weekly patients (9 of 51) achieved clinical response (EOW versus placebo strata-adjusted difference, 5.6 % [95 % CI: -4.0 % to 15.3 %]; $p = 0.25$; weekly versus placebo strata-adjusted difference, 13.7 % [CI: 1.7 % to 25.7 %]; $p = 0.025$). Serious adverse event rates were 3.9 %, 5.8 %, and 7.8 % for placebo, EOW, and weekly patients, respectively (EOW versus placebo difference, 1.8 % [CI: -6.4 % to 10.1 %]; weekly versus placebo difference, 3.9 % [CI: -5.2 % to 13.0 %]). Significantly greater improvements in patient-reported outcomes and pain were seen in the weekly dosing group than in the placebo group. A decrease in response was seen after the switch from weekly to EOW dosing in period 2. The authors concluded that adalimumab dosed once per week alleviates moderate to severe HS. Drawbacks of this study were

van Rappard et al. (2013) provided an overview of the current evidence regarding off-

label treatment of HS with TNF- α inhibitors; a systematic search was performed in MEDLINE, EMBASE and CENTRAL. Any type of original article concerning HS patients treated with infliximab, etanercept and/or adalimumab was included. No language restriction was applied. After full-text screening 65 studies involving 459 patients met the inclusion criteria and were subjected to data extraction. Four randomized controlled trials (RCTs) were available, and the remainders were case series or reports. Only RCTs were subjected to methodological quality assessment. Based on efficacy data extracted from the case reports, a moderate to good response was seen in 82 % of the patients treated with infliximab, 76 % of the patients treated with adalimumab, and 68 % of the patients treated with etanercept. Due to the moderate level of evidence only a weak recommendation can be provided. If conventional treatment options fail, the use of TNF- α inhibitors can be a useful supplement for the treatment of recurrent severe HS. Infliximab should be preferred based on the most encouraging results regarding efficacy and expenses. Also, adalimumab seems promising when administered in higher doses. The use of etanercept should be discouraged.

Blok et al. (2013) noted that HS is a difficult disease to treat. Although the pathogenesis of this inflammatory skin disease is largely unknown, the important role of the immune system has been demonstrated in both experimental and clinical studies. Clinicians are therefore increasingly prescribing systemic treatments with immunosuppressive agents, but the more traditionally used systemic retinoids, especially isotretinoin, also remain relatively common therapies. In order to provide an overview of all currently available systemic immunosuppressive agents and retinoids for the treatment of HS, a systematic search was performed using the Medline and Embase databases. All published papers concerning systemic retinoids or immunosuppressive treatments for HS in adults were included. The primary endpoints were the percentages of significant responders, moderate responders and non-responders. Other endpoints were the relapse rate and adverse events. In total, 87 papers were included, comprising 518 patients with HS who were treated with systemic retinoids, biological agents or another immunosuppressive agents, including colchicine, cyclosporine, dapsone or methotrexate. The highest response rates were observed with infliximab, adalimumab and acitretin. Overall, the quality of evidence was low and differed between the agents, making direct comparisons difficult. However, based on the amount of evidence, infliximab and adalimumab were the most effective agents. Acitretin was also effective in HS, although the quality of the evidence was low. The therapeutic effect of isotretinoin is questionable. Randomized controlled trials are needed to confirm the effectiveness of acitretin, and to identify the most effective immunosuppressive agents in HS.

Samyia and Brassard (2013) high-lighted the use of biologic agents for the treatment of recalcitrant HS. These investigators reported on a 48-year old male with a 15-year history of refractory perianal-inguinal-buttock HS who, despite receiving numerous surgical drainages and traditional medical treatment for HS, still had severe pain. After trialing etanercept and infliximab with methotrexate, the patient had marked improvement with adalimumab. A literature review of biologics therapy was also performed. After trialing many traditional therapies, these researchers found that adalimumab appears to be the most effective treatment modality for the patient. A literature search revealed 53 articles on biologics therapy in HS. These articles were summarized. The authors concluded that biologic agents have been shown to have variable results in the treatment of refractory HS. Enough low-grade evidence has been accumulated to make the use of these agents suitable in HS. Moreover, they stated that until more clinical trials are performed on this topic, physicians should use clinical judgment when treating HS with biologic agents and be cautious by watching for significant adverse effects.

Martin-Ezquerria and associates (2014) performed a retrospective study from 7 tertiary Spanish centers reviewing the charts of patients with HS treated with biological drugs. Retrieved information included epidemiological data, clinical features, pain intensity, Hurley stage, laboratory data and therapeutic outcomes. A total of 19 patients were included in the study; 10 men (52.6 %) and 9 women. Eight patients (42 %) showed a Hurley severity stage II and 11 a stage III (57.8 %). Adalimumab was prescribed as the first biological treatment in 9 out of 19 cases (47.3 %), whereas infliximab was prescribed in 7 cases (36.8 %), ustekinumab in 2 cases (10.5 %) and etanercept in 1 (5.2 %). A complete response was observed in 3 patients (2 cases with infliximab and 1 case with ustekinumab), a partial improvement in 10 patients and in 6 patients no clinical improvement was noted. One patient referred worsening of the skin symptoms. In 6 cases, a 2nd biological treatment was prescribed. In 3 of such cases, a partial improvement was noted, whereas in 3 cases no clinical improvement was observed. In 2 cases a switch to a

3rd biological drug was indicated, with a partial improvement in 1 case. The authors concluded that biological drugs could be a potential and effective therapeutic option for patients with severe HS. Complete and persistent clinical responses were rarely obtained (15 %) and partial responses were achieved in approximately 50 % of patients. No specific markers for a therapeutic response have been identified. No definitive conclusions regarding the most effective biological drug for HS could be drawn. Higher dosage schedules seem to be associated with higher response rates. The lack of response of one particular drug does not preclude a potential efficacy to another biological treatment.

An UpToDate review on “Treatment of hidradenitis suppurativa. (acne inversa)” (Margesson and Danby, 2014) states that “The main pharmacologic agents utilized for the management of severe and refractory disease are biologic therapies and oral retinoids. Based upon the data in support of its efficacy, infliximab is considered a first-line pharmacologic agent for the treatment of severe and refractory HS/AI. However, the requirement for intravenous administration, the presence of patient-specific contraindications, and the expense of the drug prohibits treatment in some patients. Alternatives to infliximab include adalimumab and acitretin. However, the data in support of these agents are less robust than for infliximab therapy. Ustekinumab, a newer biologic agent, has demonstrated benefit in small numbers of patients”.

Hunter Syndrome (Mucopolysaccharidosis Type II)

An UpToDate review on “Mucopolysaccharidoses: Complications and management” (Wynn, 2017) does not mention adalimumab as a therapeutic option.

Hurler Syndrome (Mucopolysaccharidosis Type I)

Polgreen et al (2017) stated that mucopolysaccharidosis I and II are lysosomal storage disorders that, despite treatment with hematopoietic cell transplantation (HCT) and/or enzyme replacement therapy (ERT), continue to cause significant skeletal abnormalities leading to pain, stiffness, physical dysfunction, and short stature. Tumor necrosis factor- α (TNF- α) is elevated in individuals with MPS I and II and associated with pain and physical dysfunction. In a 32-week, randomized, double-blind, placebo-controlled, cross-over, pilot study, these investigators examined the safety and effectiveness of adalimumab in patients with MPS I and II at a dose of 20 mg (weight 15 to less than 30 kg) or 40 mg (weight 30 kg or greater) administered subcutaneously every other week or saline placebo for 16 weeks. Subjects were examined at baseline, week 16, and week 32 with the Children's Health Questionnaire – Parent Form 50 (CHQ-PF50), the Pediatric Pain Questionnaire (PPQ), range-of-motion (ROM) measurements, anthropometry, 6-min walk test (6MWT), hand dynamometer, and laboratory evaluations for safety. The primary outcome was safety and primary effectiveness outcome was bodily pain (BP) measured by the CHQ-PF50. Two subjects, 1 with MPS I and 1 with MPS II, completed the study. Adalimumab was well-tolerated and there were no serious AEs. Standardized BP scores for age and gender were higher (i.e., less pain) at the end of the treatment versus placebo phase for both subjects. Subject #1 became unblinded during treatment due to skin erythema. Behavior measured by both CHQ-PF50; and parental report improved during treatment compared to placebo in both subjects. ROM improved by greater than 5° in 7 of 8 joints in Subject #1 and 5 of 8 joints in Subject #2 (range of 7.0° to 52.8°). There was no change in the PPQ, 6MWT, or hand dynamometer. The authors concluded that findings from this small pilot study suggested that treatment with adalimumab was safe, tolerable, and may improve ROM, physical function, and possibly pain, in children with MPS I or II. Moreover, these researchers stated that additional clinical trials are needed before this therapy should be recommended as part of clinical care.

Furthermore, an UpToDate review on “Mucopolysaccharidoses: Treatment” (Hahn, 2023) does not mention adalimumab as a management / therapeutic option.

Juvenile Idiopathic Arthritis

Adalimumab has been approved by the FDA for reducing signs and symptoms of moderately to severely active polyarticular juvenile idiopathic arthritis (JIA) in patients 4 years of age and older. In the United States, JIA is commonly referred to as juvenile rheumatoid arthritis (JRA). JIA is the most common chronic rheumatic disease in children with onset before age 16. Typical symptoms include stiffness when awakening, limping, and joint swelling. Any joint can be affected, and inflammation may limit the mobility of the affected joints. Polyarticular JIA is a form of arthritis affecting 5 or more joints,

usually in a symmetrical fashion. While it was once believed that most children eventually outgrow JIA, it is now known that between 25 and 70 % of children with JIA will still have active disease into adulthood.

The approval of adalimumab for JIA was based on the results of a 48-week study and a subsequent open-label extension evaluating the efficacy and safety of adalimumab in children with JIA. The 48-week phase III study included 171 children (4 to 17 years of age) with polyarticular JIA. In the first part of this study, 2 groups of children -- those taking methotrexate (MTX) and those not taking MTX -- received open-label adalimumab (up to a maximum of 40 mg) every other week for 16 weeks. Patient responses were measured using the American College of Rheumatology Pediatric (ACR Pedi) 30 score, which represents a 30 % or greater improvement in JIA signs and symptoms. Children who showed a positive clinical response (n = 133) entered the second part of the study and were randomized to receive adalimumab or placebo for an additional 32 weeks or until disease flare. A flare was defined as a worsening of 30 % or more in at least 3 of the 6 ACR Pedi response variables, a minimum of 2 active joints, and no more than 1 indicator improving by 30 %. In the second part of this study, significantly fewer children receiving adalimumab demonstrated disease flare compared to children on placebo, both without MTX (43 % versus 71 %) and with MTX (37 % versus 65 %). Additionally, more children treated with adalimumab continued to show ACR Pedi 30/50/70 responses at week 48 compared to placebo. At the conclusion of the 48-week study or at the time of disease flare during the double-blind phase, children could enter the open-label extension period. Efficacy and safety were assessed at routine intervals throughout the study.

ACR Pedi responses were maintained for up to 2 years in children who received adalimumab throughout the study. Upon initiation of treatment with adalimumab, the most common adverse reactions that occurred were injection site pain and injection site reaction (19 % and 16 %, respectively). In general, adverse reactions in children were similar in frequency and type to those seen in adult patients. Severe adverse reactions reported in the clinical trial in JIA included neutropenia, streptococcal pharyngitis, increased aminotransferases, herpes zoster, myositis, metrorrhagia and appendicitis. Serious infections were observed in 4 % of children within approximately 2 years of initiation of treatment with adalimumab and included cases of herpes simplex, pneumonia, urinary tract infection, pharyngitis, and herpes zoster. Recommended dosing in JIA is based upon weight. For children 15 kg (33 lbs) to less than 30 kg (66 lbs), recommended dose is 20 mg by injection every other week. For children 30 kg (66 lbs) and greater, recommended dose is 40 mg by injection every other week.

Lichen Planus

Chao (2009) noted that lichen planus (LP) is a common, chronic, inflammatory dermatosis that may involve the skin as well as oral and genital mucosa. Characterized by distinctive purplish papules often featuring white reticular scale, LP commonly is resistant to treatment. In this single-case study, subject presented with extensive, violaceous, and lace-like whitish lesions on the distal extremities, including the hands and feet, and the vulva. About 10 % to 12 % of her body surface area (BSA) was involved, and her condition became progressively worse over time, with thick plaques developing on the buccal mucosa and tongue. After several conventional therapies failed, subject underwent treatment with adalimumab. An almost clear response was noted by week 6, and the patient's lesions remained almost fully resolved after week 22. The authors concluded that additional studies are needed to examine the safety and effectiveness of adalimumab for the treatment of LP.

Didona et al (2022) stated that oral LP (OLP) is a chronic inflammatory disease of the oral mucosa. Several clinical subtypes of OLP have been reported, including the reticular and erosive one. On the one hand, reticular OLP is usually asymptomatic and is characterized by white streaks surrounded by well-defined erythematous borders. On the other hand, erosive OLP shows ulcerations and erosions surrounded by erythematous mucosa. While reticular OLP is relatively easy to control, erosive OLP is extremely painful and refractory to therapies, limiting the QoL of the patients. Furthermore, treating erosive OLP is extremely tricky, and a gold standard treatment has not yet been established. However, several therapeutic approaches have been reported as effective, including systemic corticosteroids, systemic retinoids, and anti-interleukin (IL)-17/anti-IL-23 drugs. The authors noted that several biologic therapies have been used in patients with refractory OLP, including anti-CD2, anti-TNF-alpha, anti-IL2, anti-IL17, anti-IL12/23, and anti-IL23 drugs. Several cases of OLP treated with anti-TNF-alpha agents have been reported. An improvement of erosive OLP was reported in a patient treated with etanercept. In addition, a patient with a severe orogenital LP was

successfully treated with infliximab and 2 other patients with a severe orogenital involvement were treated with adalimumab. However, emerging data suggest that TNF-alpha inhibitors may trigger OLP.

Chen et al (2022) noted that the use of TNF- α inhibitors for the management of LP has been reported in the literature. Chao et al reported that adalimumab led to the remission of cutaneous and mucosal LP. However, there is also some evidence that TNF- α inhibitors can paradoxically cause LP-like eruptions.

Furthermore, UpToDate reviews on "Lichen planus" (Goldstein et al, 2023) and "Vulvar lichen planus" (Copper and Arnold, 2023) do not mention adalimumab as a management/therapeutic option.

Lumbar Disc Herniation

In a multi-center, double-blind, randomized, controlled trial, Genevay and colleagues (2010) evaluated the effectiveness of adalimumab in patients with radicular pain due to lumbar disc herniation. Patients with acute (less than 12 weeks) and severe (Oswestry Disability index greater than 50) radicular leg pain and imaging-confirmed lumbar disc herniation were randomized to receive as adjuvant therapy either 2 subcutaneous injections of adalimumab (40 mg) at 7-day interval or matching placebo. The primary outcome was leg pain, which was recorded every day for 10 days and at 6 weeks and 6 months based on a visual analog scale (0 to 100). Of the 265 patients screened, 61 were enrolled (adalimumab = 31) and 4 were lost to follow-up. Over time, the evolution of leg pain was more favorable in the adalimumab group than in the placebo group ($p < 0.001$). However, the effect size was relatively small and at last follow-up the difference was 13.8 (95 % confidence interval: -11.5 to 39.0). In the adalimumab group, twice as many patients fulfilled the criteria for "responders" and for "low residual disease impact" ($p < 0.05$) and fewer surgical discectomies were performed (6 versus 13, $p = 0.04$). The authors concluded that the addition of a short course of adalimumab to the treatment regimen of patients suffering from acute and severe sciatica resulted in a small decrease in leg pain and in significantly fewer surgical procedures. Limitations of this study include small sample sizes, inability to determine the sufficient dose and best route of administration for TNF-alpha inhibitors in radiculopathy because of disk herniation, or to identify subgroups of patients most likely to respond to this treatment. More studies are needed to ascertain the adequate dose and the best route of administration of TNF-alpha inhibitors, elucidating the effective therapeutic role of TNF-alpha inhibitors in this disease and determining the dosage, times and route of administration, as well as evaluating the possible occurrence of harmful events, as serious infections or adverse cardiovascular events, as seen in patients with rheumatoid arthritis treated with biological drugs.

Osteoarthritis

In a pilot study ($n = 12$), Magnano et al. (2007) examined if adalimumab can safely improve symptoms of erosive/inflammatory osteoarthritis (EOA). Patients greater than 45 years of age with EOA of the hands defined by greater than or equal to 2 tender and greater than or equal to 2 swollen joints (distal inter-phalangeal, proximal inter-phalangeal, first carpo-metacarpal) despite NSAID therapy were eligible. Patients were excluded for autoimmune arthritis, recent disease modifying anti-rheumatic drug use, prior use of anti-TNF therapy, infection, malignancy, or poorly controlled medical conditions. All patients received adalimumab 40 mg every other week for 12 weeks. Safety was assessed 4 weeks after the final dose. Primary endpoints included safety and ACR response. Patients were predominantly female with a mean age of 60 years and 12 years of arthritis. All subjects completed the study and safety follow-up. Adverse events were mild without necessitating discontinuation of study drug. After 12 weeks, there was a statistically significant improvement in the number of swollen joints compared to baseline ($p < 0.01$). One patient achieved an ACR20 response and 42 % achieved an OMERACT-OARSI response. Although these investigators detected no statistically significant improvement in the number of tender joints, grip strength, disability, pain, or global disease assessments, trends suggested modest improvement in all efficacy measures. The authors concluded that the findings of this small open-label study of patients with EOA demonstrated that adalimumab was well-tolerated. Treatment with adalimumab for 3 months did not significantly improve the signs and symptoms of EOA and most patients did not achieve an ACR20. Trends suggested improvement and individual patients had some benefit. Factors limiting interpretation of this study include the lack of a control group, outcomes chosen, number of patients treated, and the duration of

treatment.

In a randomized, double-blind, placebo-controlled, parallel group, multi-center study, Chevalier et al (2015) examined the effectiveness of adalimumab in patients with painful refractory (non-responders to analgesics and NSAIDs) hand OA. Patients were randomized to 1/1 adalimumab 40 mg for 2 subcutaneous injections at a 15-day interval or placebo and monitored for 6 months. The primary outcome was the percentage of patients with an improvement of more than 50 % in global pain (visual analog scale) between week 0 (W0) and week 6 (W6). Secondary outcomes included the number of painful joints, swollen joints, morning stiffness duration, patient and practitioner global assessments, functional indexes for hand OA, and consumption of analgesics. Analysis on the mean primary outcome measure was done on patients who received at least 1 injection. A total of 99 patients were recruited and 85 patients were randomized. Among them, 37 patients in the placebo group and 41 in the adalimumab group received at least 1 injection and were evaluated at W6 (n = 78) on the main effectiveness outcome. Mean age was 62 years, 85 % were women, and mean level of pain was 62 mm at W0. At W6, 35.1 % in the adalimumab group versus 27.3 % in the placebo group had a pain reduction greater than or equal to 50 % (RR 1.12 (95 % CI: 0.82 to 1.54; p = 0.48). There were no statistical differences for all secondary end-points. The rate of adverse events was similar in the 2 groups. The authors concluded that adalimumab was not superior to placebo to alleviate pain in patients with hand OA not responding to analgesics and NSAIDs.

Pediatric Crohn's Disease

Adalimumab has been used successfully in children and adolescents with Crohn's disease who do not respond or develop infusion reactions to infliximab. A randomized controlled trial (Hyams, et al., 2012) comparing two doses of adalimumab in 188 patients with refractory moderate-to-severe pediatric Crohn's disease found comparable rates of remission with the two dosing regimens, occurring in about one-third of subjects at six months. In a retrospective series of 115 persons with refractory pediatric Crohn's disease treated with adalimumab (Rosh, et al., 2009), 70 percent had a clinical response at one year.

In September 2014, the FDA expanded approval of adalimumab for reducing signs and symptoms, and achieving and maintaining clinical remission, in pediatric Crohn's disease patients 6 years of age and older when certain other treatments have not worked well enough (Abbvie, 2014). The FDA approval of adalimumab for pediatric Crohn's was supported by the Phase 3 IMaGINE-1 trial, a multi-center, randomized, double blind trial, that evaluated multiple dosing strategies of adalimumab to induce and maintain clinical remission in pediatric patients 6 to 17 years of age with moderately to severely active Crohn's disease for whom certain other treatments have not worked well enough (Hyams, et al., 2012).

Hyams et al. (2012) evaluated the safety and effectiveness of adalimumab double-blind maintenance dosing regimens following open-label induction for pediatric patients with moderate-to-severe Crohn's disease (CD). These investigators studied 192 patients with Pediatric Crohn's Disease Activity Index (PCDAI) scores greater than 30 for whom conventional treatment was unsuccessful. Patients received open-label induction therapy with subcutaneous adalimumab at weeks 0 and 2 (160 mg and 80 mg, or 80 mg and 40 mg, for body weight [BW] greater than or equal to 40 kg or less than 40kg). At week 4, 188 patients were assigned to groups based on achievement of clinical response (defined as PCDAI decrease less than or equal to 15 points from baseline, 155/188 [82.4 %]) and prior exposure to infliximab (82/188, 43.6 %). Groups were given double-blind maintenance therapy with adalimumab at high (40 mg or 20 mg for BW greater than or equal to 40 kg or less than 40 kg; n = 93) or low doses (20 mg or 10 mg for BW greater than or equal to 40 kg or less than 40 kg; n = 95), every other week for 48 weeks. Clinical remission (PCDAI less than or equal to 10) at week 26 (the primary endpoint) was compared between groups using the Cochran-Mantel-Haenszel test, adjusting for strata, with non-responder imputation. Adverse events were monitored to evaluate safety; 152/188 patients (80.9 %) completed all 26 weeks of the study. At week 26, 63 patients (33.5 %) were in clinical remission, with no significant difference between high- and low-dose groups (36/93, 38.7 % versus 27/95, 28.4 %; p = 0.075). No new safety signals were detected. The authors concluded that adalimumab induced and maintained clinical remission of children with CD, with a safety profile comparable to of adult patients with CD. More children that received high than low dose were in remission at week 26, but the difference between dose groups was not statistically significant.

Plaque Psoriasis

Adalimumab has been approved by the FDA for the treatment for adults with moderate to severe chronic plaque psoriasis who are candidates for systemic therapy or phototherapy, and when other systemic therapies are medically less appropriate. According to the FDA-approved labeling, adalimumab should only be administered to patients who will be closely monitored and have regular follow-up visits with a physician. Chronic plaque psoriasis is an autoimmune disease characterized by inflamed, scaly skin lesions known as plaques, which may crack and bleed. While psoriasis can occur in people of all ages, it typically appears in patients between the ages of 15 and 35 years. Approximately 25 % of persons with chronic plaque psoriasis exhibit moderate to severe disease. Up to 30 % of psoriasis patients develop psoriatic arthritis. Treatment may include topical agents, phototherapy or oral or injectable medications.

The FDA approval of adalimumab for chronic plaque psoriasis was based on 2 pivotal trials, REVEAL and CHAMPION, showing that approximately 3 out of 4 patients achieved 75 % clearance or better at week 16 of treatment versus placebo. Both studies evaluated the efficacy and safety of HUMIRA in clearing skin in moderate to severe adult plaque psoriasis patients versus placebo. In addition, CHAMPION compared a biologic medication to methotrexate, a standard systemic treatment for psoriasis. In each trial, reduction in disease activity was determined by the Psoriasis Area and Severity Index (PASI) and Physician's Global Assessment (PGA). In REVEAL, a 52-week trial, the short-term and sustained clinical efficacy and safety of adalimumab were evaluated in 1,212 patients with moderate to severe chronic plaque psoriasis. Patients experienced a significant reduction in the signs and symptoms of their disease at 16 weeks when treated with adalimumab. Specifically, 71 % of patients receiving adalimumab achieved PASI 75 compared to 7 % of patients receiving placebo at week 16. At week 16, 62 % of adalimumab-treated patients achieved a PGA score of clear or minimal compared to 4 % of placebo-treated patients. In CHAMPION, a 16-week study evaluating 271 psoriasis patients, adalimumab-treated patients experienced a significant reduction in the signs and symptoms of their disease compared with methotrexate or placebo-treated patients. Seventy-eight percent of patients treated with adalimumab (n = 99) achieved a PASI 75 response, compared to 19 % of patients treated with placebo (n = 48). Seventy-one percent of patients treated with adalimumab achieved a PGA score of clear or minimal at 16 weeks of treatment, compared with 10 % of placebo-treated patients. The safety profile of adalimumab in the plaque psoriasis clinical trials was reported to be similar to that seen in adalimumab clinical trials for rheumatoid arthritis. The most commonly reported adverse events in adalimumab psoriasis trials were upper respiratory tract infection, nasopharyngitis (inflammation of the nose and pharynx), headache, sinusitis and arthralgia. In clinical studies of plaque psoriasis, patients are treated with an initial 80-mg dose of adalimumab (2 40-mg injections) followed by 1 adalimumab injection (40 mg) 1 week later. After that, a maintenance dose of 40 mg was administered every other week.

Psoriatic Arthritis

The FDA has approved adalimumab for reducing signs and symptoms of active arthritis in patients with psoriatic arthritis. The FDA-approved product labeling for Humira states that adalimumab can be used alone or in combination with methotrexate or with DMARDs. The FDA approval of Humira was based on 2 multi-center randomized controlled clinical studies evaluating the safety and efficacy of adalimumab compared with placebo in 413 patients with moderate to severely active psoriatic arthritis (greater than 3 swollen and greater than 3 tender joints) who have had an inadequate response to non-steroidal anti-inflammatory drugs (NSAIDs). In 1 study (Mease et al., 2005), 313 patients with moderately to severely active psoriatic arthritis and a history of inadequate response to NSAIDs were randomized to receive 40 mg adalimumab or placebo subcutaneously every other week for 24 weeks. Study participants had the following forms of psoriatic arthritis:

Patients on methotrexate therapy (158 of 313 patients) at enrollment (stable dose of less than or equal to 30 mg/week for greater than 1 month) could continue methotrexate at the same dose. Study visits were at baseline, weeks 2 and 4, and every 4 weeks thereafter. The primary efficacy end points were the ACR 20 % improvement (ACR20) response at week 12 and the change in the modified total Sharp score of structural damage at week 25. Secondary end points were measures of joint disease, disability, and quality of life in all patients, as well as the severity of skin disease in those patients with psoriasis involving at least 3 % of body surface area. At week 12, 58 % of the adalimumab-treated

patients (87 of 151) achieved an ACR20 response, compared with 14 % of the placebo-treated patients (23 of 162) ($p < 0.001$). At week 24, similar ACR20 response rates were maintained and the mean change in the modified total Sharp score was -0.2 in patients receiving adalimumab and 1.0 in those receiving placebo ($p < 0.001$). Similar responses were seen in patients with each of the subtypes of psoriatic arthritis, although few patients were enrolled with the arthritis mutilans and ankylosing spondylitis-like subtypes. Responses were similar in patients who were or were not receiving concomitant methotrexate therapy at baseline. Patients with psoriatic involvement of at least 3 % body surface area (BSA) were evaluated for Psoriatic Area and Severity Index (PASI) responses. Among the 69 adalimumab-treated patients evaluated for PASI responses, 59 % achieved a 75 % PASI improvement response and 42 % achieved a 90 % PASI improvement response at 24 weeks, compared with 1 % of the 69 placebo-treated patients evaluated ($p < 0.001$). The investigators reported that disability and quality of life measures were also significantly improved with adalimumab treatment compared with placebo.

Similar results were seen in an additional, 12-week study in 100 patients with moderate to severe psoriatic arthritis who had suboptimal response to DMARD therapy as manifested by greater than or equal to 3 tender joints and greater than or equal to 3 swollen joints at enrollment.

The manufacturer reported that the rates of adverse events and serious adverse events in clinical trials of adalimumab for psoriatic arthritis submitted for FDA approval were comparable with clinical trials of adalimumab in rheumatoid arthritis. Among patients taking adalimumab, the most common adverse events (those affecting at least 5 % of patients) were upper respiratory infection, nasopharyngitis, injection site reaction, headache and hypertension. The safety profile of adalimumab in these clinical trials was similar to that observed in the clinical trials of adalimumab for rheumatoid arthritis that were submitted for FDA approval. The FDA-approved product labeling states that the safety and efficacy of adalimumab in pediatric patients has not been established.

Pyoderma Gangrenosum

Suarez-Perez et al. (2012) described the epidemiological and clinical characteristics of patients with pyoderma gangrenosum (PG) along with their experience of treating the condition in a referral hospital in Malaga, Spain. A retrospective, observational study was undertaken between January 2000 and December 2009 and included all patients diagnosed with PG. The incidence of PG in the authors' reference population is 3.26 cases per million inhabitants per year. The most frequent concomitant systemic disease was UC (5 cases, 33 %). In 4 patients with that disease, PG appeared during a flare-up. In 80 % of cases, patients were not referred to a dermatologist during the initial phase of PG, and most referrals were from gastroenterology or general surgery (4 patients each, 52 %). The authors concluded that patients with PG were often referred to dermatologists by other specialists after a varying period of time has elapsed without achieving an accurate diagnosis. In these patients, especially those between 20 and 40 years of age, it is essential to rule out concomitant disease. Adalimumab is a good treatment option for PG.

Gisondi and Girolomoni (2013) investigated the impact of TNF-alpha antagonists on health-related quality of life (HRQoL) in selected skin diseases, i.e., chronic plaque psoriasis, Behcet's disease (BD), hidradenitis suppurativa (HS) and PG (PG). These investigators carried out a systematic literature search of Medline (2000 to April 2013) using the Cochrane highly sensitive and specific search strategy. Citations were screened for randomized, controlled trials of TNF-alpha antagonists (adalimumab, etanercept and infliximab) versus placebo in adults with psoriasis, BD, HS or PG. From the literature it is evident that skin diseases can affect physical, psychological, social and occupational aspects of everyday life. Tumor necrosis factor-alpha antagonists induced consistent benefits across health outcomes in psoriasis, but only monoclonal antibodies, infliximab and adalimumab were effective in improving QoL in patients with BD, HS and PG. Dermatology Life Quality Index was the most common used tool for investigating HRQoL. For the majority of patients with skin diseases, the most important negative impacts on QoL were appearance related. Generally, the burden on QoL was correlated to the severity of skin disease and the improvement in QoL achieved by TNF-alpha blockers was proportional to the degree of disease remission. In general, achieving the highest clearing of skin disease with anti-TNF-alpha agents is required for optimal improvement in QoL.

Agarwal and Andrews (2013) provided an up-to-date review of the published treatment

effectiveness of currently available therapies for IBD-related PG in the biologic era. Systematic review of cases published post-2003 since the broad availability of anti-TNF-alpha therapy. Cases that did not have co-existent IBD, were non-English language, of pediatric age or without data on response to therapy were excluded. A total of 60 cases were identified; 55 % female, 50 % UC, 45 % CD, 5 % IBD-U. At PG diagnosis, 58 % had active and only 15 % inactive IBD, with 27 % with IBD activity unspecified. Predominant sites were lower limb (48 %) and peristomally (25 %); 42 % had multiple lesions. In 12 %, trauma preceded PG. In 42 %, new PG appeared while on IBD-specific therapy, while 28 % were on no therapy and in 30 %, IBD therapy was unspecified. Of patients on no therapy at PG onset (n = 17), 16 healed; 7 with first- and 8 with second-line therapy. In total, 34/60 patients received infliximab, 4 received adalimumab, 2 had both; with 33 (92 %) responding to one or the other. There was no correlation of PG duration or size with healing times. The authors concluded that PG appears predominantly during active IBD and is seen equally in CD and UC. New PG may be a manifestation of recrudescence IBD or it follow trauma. Anti-TNF-alpha therapy as a first-line agent for PG should be considered, as it appears to be highly effective.

Arguelles-Arias et al. (2013) examined the characteristics of PG associated with CD or UC and which treatments were prescribed in Spanish clinical practice. In this retrospective, observational study, the medical records from all patients with IBD and a diagnosis of PG attended by the gastroenterology departments of 12 Spanish hospitals were reviewed. Data on patient demographics and characteristics, underlying IBD and treatment, and PG characteristics, treatment, and outcome were collected and analyzed. The data from 67 patients were analyzed (41 [61.2 %] women, 41 [61.2 %] with CD, 25 [37.3 %] with UC, and 1 [1.5 %] with indeterminate disease). The underlying disease was in remission in approximately 1/3 of patients at the time of presentation of PG. Healing was achieved in all patients (in 3 without any systemic therapy). Oral corticosteroids were taken by 51 patients (76.1 %), almost always as first-line treatment, although definitive healing was attained in 19 (28.4 %). Biologic agents such as infliximab and adalimumab were taken by 31 patients (46.3 %) at some point (first-line in 6 patients [9.0 %]), with definitive healing in 29 patients (93.5 %). The authors concluded that oral corticosteroid therapy remains the most common treatment for PG associated with IBD. Biologic therapies such as infliximab and adalimumab should also be considered.

Marzano et al. (2013) stated that PG and Sweet's syndrome (SS) are skin diseases usually presenting with recurrent ulcers and erythematous plaques, respectively. The accumulation of neutrophils in the skin, characteristic of these conditions, led to coin the term of neutrophilic dermatoses to define them. Recently, neutrophilic dermatoses have been included in the group of auto-inflammatory diseases, which classically comprises genetically determined forms due to mutations of genes regulating the innate immune response. Both PG and SS are frequently associated with IBDs; however, IBD patients develop PG in 1 to 3 % of cases, whereas SS is rarer. Clinically, PG presents with deep erythematous-to-violaceous painful ulcers with well-defined borders; bullous, pustular, and vegetative variants can also occur. Sweet's syndrome is characterized by the abrupt onset of fever, peripheral neutrophilia, tender erythematous skin lesions, and a diffuse neutrophilic dermal infiltrate. It is also known as acute febrile neutrophilic dermatosis. Treatment of PG involves a combination of wound care, topical medications, antibiotics for secondary infections, and treatment of the underlying IBD. Topical therapies include corticosteroids and the calcineurin inhibitor tacrolimus. The most frequently used systemic medications are corticosteroids and cyclosporine, in monotherapy or in combination. Dapsone, azathioprine, cyclophosphamide, methotrexate, intravenous immunoglobulins, mycophenolate mofetil, and plasmapheresis are considered second-line agents. Hyperbaric oxygen, as supportive therapy, can be added. Anti-TNF-alpha agents such as etanercept, infliximab, and adalimumab are used in refractory cases. Sweet's syndrome is usually responsive to oral corticosteroids, and the above-mentioned immunosuppressants should be considered in resistant or highly relapsing cases.

Also, an UpToDate review on "Pyoderma gangrenosum: Treatment and prognosis" (Schadt, 2014) states that "Second line and adjunctive therapies -- In addition to infliximab, other biologic TNF-alpha inhibitors may be beneficial in PG. Adalimumab (40 mg weekly, 40 mg twice monthly, and other regimens) has been associated with ulcer healing in case reports. Most of the published cases have involved patients with concomitant inflammatory bowel disease or rheumatoid arthritis. Improvement in PG with etanercept (25 to 50 mg twice weekly) has also been reported in a small retrospective series and case report. However, in our experience, adalimumab seems to be more efficacious".

Reiter's Syndrome (Reactive Arthritis)

A Medscape review on “Reactive Arthritis” (last reviewed October 31, 2015) did not mention adalimumab as a therapeutic option.

Furthermore, an UpToDate review on “Reactive arthritis” (Yu, 2017) states that “An observational study in France reported the efficacy and safety of anti-TNF therapies in 10 patients with reactive arthritis of recent onset (within 12 months). Five, four, and one patient(s) each received infliximab, etanercept, and adalimumab, respectively. Nine of 10 patients met response criteria, with reduction in tender and swollen joint counts, pain score, and serum C-reactive protein (CRP) levels. All eight patients taking glucocorticoids successfully discontinued them after a median of four months. Six patients discontinued the anti-TNF therapy after a median of 7.5 months; three relapsed after stopping but responded to retreatment”.

Rheumatoid Arthritis

Adalimumab has been approved by the FDA for reducing signs and symptoms and inhibiting the progression of structural damage in adult patients with moderately to severely active rheumatoid arthritis. Adalimumab can be used alone or in combination with methotrexate or other non-biologic disease modifying anti-rheumatic drugs (DMARDs). The efficacy and safety of adalimumab were assessed in 5 randomized, double-blind studies in 2,869 subjects aged 18 years and older with active rheumatoid arthritis diagnosed according to American College of Rheumatology (ACR) criteria. All subjects had at least 6 tender and 9 swollen joints. Humira was administered subcutaneously in combination with methotrexate or as monotherapy or with other DMARDs. Two studies involved 815 patients who had failed to respond to DMARDs; 1 study involved 619 patients who had an inadequate response to methotrexate. One study assessed 636 patients who were either DMARD-naïve or were permitted to remain on their pre-existing rheumatologic therapy. One study evaluated 799 patients with moderately to severely active rheumatoid arthritis of less than 3 years duration who were methotrexate naïve. In all 5 studies, adalimumab showed significantly greater improvement than placebo in standardized indices of disability and health outcomes from baseline to the end of study. One of the 5 studies also examined the effect adalimumab on inhibition of disease progression, as detected by X-ray results. Subjects treated with adalimumab and methotrexate showed significantly less radiological progression of disease than subjects treated with methotrexate alone. Subjects treated with adalimumab and methotrexate showed significantly less radiological progression of disease than subjects treated with methotrexate alone.

According to guidelines from the ACR (Saag et al., 2008), patients with early rheumatology with low or moderate disease activity (in study) were not considered candidates for biologic therapy. The use of anti-tumor necrosis factor (TNF) agent in combination with methotrexate was recommended if high disease activity was present for less than 3 months with features of a poor prognosis.

The American College of Rheumatology (ACR) conducted a systematic review to synthesize the evidence for the benefits and harms of various treatment options. Their goal was to develop evidence-based, pharmacologic treatment guideline for rheumatoid arthritis. The *2015 American College of Rheumatology Guidelines for the Treatment of Rheumatoid Arthritis* provided “strong” recommendations for established RA and symptomatic early RA.

For established RA, the guidelines state “if the disease activity is low, in patients who have never taken a DMARD, the recommendation is to use DMARD monotherapy (methotrexate preferred) over TNFi”. “If disease activity remains moderate or high despite DMARD monotherapy, the recommendation is to use combination traditional [conventional] DMARDs or add a TNFi or a non-TNF biologic or tofacitinib (all choices with or without methotrexate, in no particular order of preference), rather than continuing DMARD monotherapy alone”. Recommendations for patients with symptomatic early RA state that “if disease activity is low, in patients who have never taken a DMARD, use DMARD monotherapy (methotrexate preferred) over double or triple therapy”. “If disease activity remains moderate or high despite DMARD monotherapy (with or without glucocorticoids), use combination DMARDs or a TNFi or a non-TNF biologic (all choices with or without methotrexate, in no particular order of preference), rather than continuing DMARD monotherapy alone”. A strong recommendation means that the panel was confident that the desirable effects of following the recommendation outweigh the undesirable effects (or vice versa), so the course of action would apply to most patients, and only a small proportion would not want to follow the recommendation (Singh et al., 2016).

The 2022 EULAR (European Alliance of Associations for Rheumatology) guidelines for RA management emphasize a treat-to-target approach, aiming for sustained remission or low

disease activity through early intervention, regular monitoring, and shared decision-making. Recommendations include use of methotrexate (MTX) as the preferred first-line therapy, typically initiated at 15 mg/week and escalated up to 25-30 mg/week as tolerated, with folic acid supplementation to reduce side effects. If MTX is contraindicated or not tolerated, alternatives such as leflunomide, sulfasalazine, or hydroxychloroquine can be considered. Hydroxychloroquine is particularly appropriate for patients with mild disease or when used in combination with other DMARDs, but is not recommended as monotherapy in individuals with poor prognostic indicators due to its limited efficacy in preventing joint damage. If treatment targets are not met within 3 to 6 months, escalation to a biologic DMARD (bDMARD) or targeted synthetic DMARD (tsDMARD) (e.g., tofacitinib, upadacitinib) is advised. Among bDMARDs, TNF inhibitors such as adalimumab are recommended, typically in combination with MTX but may be used as monotherapy when MTX is unsuitable. The guidelines also stress the importance of considering comorbidities and safety profiles when selecting and escalating therapy (Smolen et al., 2023).

Sarcoidosis

Cutaneous sarcoidosis may be a chronic disease with important morbidity requiring aggressive therapy. The effectiveness of different anti-TNF-alpha treatments in refractory cutaneous and systemic sarcoidosis has been reported previously. Thielen et al (2009) reported the first patient with chronic cutaneous sarcoidosis who responded to dose escalation of anti-TNF-alpha agents that have been ineffective at the standard dosage, illustrating that the optimal dosing regimen has still to be defined for this indication before considering difficult-to-treat patients as non-responders. This case report also illustrated that the fusion protein etanercept, even used at a high dosage, may be less effective for the treatment of cutaneous sarcoidosis than the monoclonal antibodies infliximab and adalimumab.

Patel (2009) stated that sarcoidosis is an inflammatory disorder characterized by the presence of non-caseating granulomas in affected organs. The presence of CD4-positive T lymphocytes and macrophages in affected organs suggests an ongoing immune response. Systemic corticosteroids remain the mainstay of treatment, but therapy is often limited by adverse effects. This is the first report of the use of adalimumab in a patient with systemic sarcoidosis with bone marrow involvement. The patient was a 42-year old African-American man with a medical history significant for hypertension and diabetes mellitus presented with anemia and thrombocytopenia of 2-month duration. He underwent physical examination, bone marrow aspiration and biopsy, chest X-ray, acid-fast bacilli stain, computed tomography with contrast, and additional laboratory tests; and was diagnosed with systemic sarcoidosis with splenomegaly and bone marrow involvement. Drug therapy included prednisone, which had to be discontinued owing to adverse effects, and adalimumab. The author concluded that this is the first report describing the use of adalimumab in a patient with systemic sarcoidosis with bone marrow involvement. Tumor necrosis factor antagonism with adalimumab was effective and well-tolerated in this patient and may be considered as a treatment option for similar cases.

Antoniou (2010) noted that sarcoidosis is a granulomatous disease with various organ manifestations in which TNF-alpha has been demonstrated to play a major pathogenic role. Conventional therapies are not always able to minimize TNF-alpha-driven inflammation and other approaches should be used. The author reviewed TNF-alpha roles in sarcoid inflammation and granuloma formation based on the literature published in the last 20 years and the therapies able to target it specifically or non-specifically in sarcoidosis were discussed. In some subsets of sarcoidosis with more rapid progression and/or therapeutic refractoriness TNF-alpha plays a more prominent role in disease pathogenesis, and its blockade might represent an appropriate therapeutic approach.

Takayasu's Arteritis

Tato and colleagues (2005) noted that treatment of patients with Takayasu's arteritis remains a demanding challenge to clinicians. In many patients, the course of the disease is characterized by frequent relapses and disease progression under conventional treatment with glucocorticoids and cytotoxic drugs. These researchers presented the case of a young woman with severe cerebrovascular and aortic involvement, who experienced disease progression in spite of more than 2 years of treatment with high doses of prednisone, methotrexate and cyclophosphamide. In this patient, treatment with adalimumab achieved clinical remission and allowed tapering of prednisone within a few months. The present case, as well as previous reports on the use of infliximab in giant

cell and Takayasu's arteries, suggested that TNFalpha-blockade may be a new, promising treatment for glucocorticoid-refractory large-vessel vasculitis.

Ulcerative Colitis

On September 28, 2012, the FDA approved adalimumab for the treatment of moderate-to-severe ulcerative colitis in adults. The approval was based on findings of 2 clinical studies showing the safety and effectiveness of adalimumab for ulcerative colitis. A total of 908 patients who had never been treated with a TNF-blocker, or who lost response to or were intolerant to TNF-blockers participated in the studies. All patients enrolled in the studies had a Mayo score of 6 to 12 and an endoscopy subscore of 2 to 3. Patients were randomly assigned to take adalimumab or a placebo. The studies were designed to measure the percentage of patients whose Mayo score decreased to 2 or less with no individual subscore of more than 1 after 8 weeks of treatment. Patients who obtained such reductions in the Mayo score were determined to have achieved clinical remission. Results from both studies showed 16.5 % to 18.5 % of patients treated with adalimumab achieved clinical remission compared with 9.2 % to 9.3 % of patients receiving placebo. Additionally, in the 2nd study, 8.5 % of patients treated with adalimumab sustained clinical remission compared with 4.1 % of patients treated with placebo. The effectiveness of adalimumab has not been established in patients with ulcerative colitis who have lost response to or were intolerant to TNF blockers.

The FDA-approved dosing regimen for adalimumab for ulcerative colitis begins with an initial dose of 160 milligrams, a 2nd dose 2 weeks later of 80 mg, and a maintenance dose of 40 mg every other week, thereafter. The drug should only continue to be used in patients who have shown evidence of clinical remission by 8 weeks of therapy.

In an open-label study, Peyrin-Biroulet et al. (2007) assessed the effectiveness of adalimumab induction therapy in patients with UC who previously responded to infliximab and then lost response or became intolerant. A total of 10 patients with UC were enrolled in a 4-week trial. Patients received a loading dose of 160 mg adalimumab at week 0 followed by 80 mg at week 2. The primary outcome measure was clinical improvement at week 4, as defined by a decrease in clinical activity index (CAI) of more than 4. Four of 10 patients (40 %) benefited from subsequent adalimumab therapy; 1 patient achieved remission (CAI less than 4) and 3 had clinical improvement at week 4; 6 patients had no response (60 %); 2 of 6 (33.3 %) subsequently underwent colectomy. This was accompanied by a decrease in median C-reactive protein (CRP) concentration from 16.8 mg/ml at baseline to 3.85 mg/ml at week 4, excluding 2 patients who underwent colectomy after 2 infusions of adalimumab. Among the 6 patients with severe colitis (CAI greater than 12) at baseline, none achieved remission and only 1 patient had clinical improvement at week 4. The authors concluded that the small advantage of adalimumab in patients with mild to moderate UC and lost response or intolerance to infliximab needs to be confirmed in randomized, double-blind, placebo-controlled trials. Furthermore, in a Cochrane review on TNF-alpha blocking agents for induction of remission in UC, Lawson et al (2006) did not mention the use of adalimumab.

Fiorino et al. (2011) summarized available data on the safety and effectiveness profile of adalimumab in patients with UC. The authors concluded that adalimumab may be effective and well-tolerated in UC. Moreover, they stated that its effectiveness in maintaining clinical remission needs to be confirmed in a randomized controlled trial.

The 2019 American College of Gastroenterology Ulcerative Colitis guidelines state there are several reasons for low serum levels of drug, including increased clearance due to increased inflammatory burden, protein loss from a permeable inflamed mucosa, the development of neutralizing antidrug antibodies, or other patient-related factors such as increased body mass index or male sex. Therefore, the approach to a patient with inadequate primary response or secondary loss of response should include careful clinical evaluation, confirmation of inflammation using objective measures (endoscopy or surrogates such as CRP or FC), exclusion of enteric infections, and assessment of serum drug concentration to address the specific contributing factors and make a decision regarding treatment options, pharmacokinetic manipulation, or cycling/swapping therapies or mechanisms. The guidelines summarize that in patients with moderately to severely active UC who are responders to anti-TNF therapy and now losing response, we suggest measuring serum drug levels and antibodies (if there is not a therapeutic level) to assess the reason for loss of response (conditional recommendation, very low quality of evidence) (Rubin 2019).

Singh, et al. (2024) state that the American Gastroenterological Association (AGA) living

guideline aims to assist healthcare practitioners in the pharmacological management of moderate-to-severe ulcerative colitis (UC). A multidisciplinary panel of experts employed the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework to prioritize clinical questions, identify patient-centered outcomes, synthesize evidence, and formulate recommendations. The panel issued 14 recommendations, endorsing the use of various advanced therapies, including infliximab, golimumab, and vedolizumab, while suggesting adalimumab, filgotinib, and mirikizumab as alternatives to no treatment. Specifically, for patients who are naïve to advanced therapies, the AGA recommends using higher-efficacy medications over lower-efficacy options like adalimumab. In patients who have previously been treated with advanced therapies, particularly tumor necrosis factor (TNF)- α antagonists, the guideline suggests using higher- or intermediate-efficacy medications instead of adalimumab. However, the AGA does recommend the use of adalimumab in combination with an immunomodulator for adult outpatients with moderate-to-severe UC, as this combination is preferred over monotherapy. The AGA also advises against thiopurine and methotrexate monotherapy for remission induction or maintenance, while emphasizing the importance of early intervention with advanced therapies rather than a gradual step-up approach after the failure of 5-aminosalicylates. Overall, this guideline provides a comprehensive, patient-centered framework for managing moderate-to-severe UC, highlighting key implementation considerations and identifying areas for future research.

Uveitis

Adalimumab (Humira) has been approved by the FDA for the treatment of non-infectious intermediate, posterior and panuveitis in adults.

In a prospective case series, Díaz-Llopis et al. (2012) evaluated adalimumab therapy in refractory uveitis. A total of 131 patients with refractory uveitis and intolerance or failure to respond to prednisone and at least 1 other systemic immunosuppressive drug participated in this study. Patients received a 40-mg adalimumab subcutaneous injection every other week for 6 months. The associated immunosuppressants were tapered after administering 3 adalimumab injections (week 6). Main outcome measures were degree of anterior and posterior chamber inflammation (Standardization of Uveitis Nomenclature Working Group criteria), immunosuppression load (as defined by Nussenblatt et al), VA (logarithm of the minimal angle of resolution [logMAR]), and macular thickness (OCT). There were 61 men and 70 women (mean age of 27.3 years). The most common causes were juvenile idiopathic arthritis in 39 patients, pars planitis in 16 patients, and Behçet's disease in 13 patients. Twenty-seven patients had uveitis of idiopathic origin. Inflammation in the anterior chamber was present in 82 % of patients and in the vitreous cavity in 59 % of patients. Anterior chamber inflammation and vitreous inflammation decreased significantly ($p < 0.001$) from a mean of 1.51 and 1.03 at baseline to 0.25 and 0.14, respectively, at 6 months. Macular thickness was 296 (102) μ at baseline versus 240 (36) μ at the 6-month visit ($p < 0.001$). Visual acuity improved by -0.3 logMAR in 32 of 150 eyes (21.3 %) and worsened by +0.3 logMAR (-15 letters) in 5 eyes (3.3 %). The dose of corticosteroids also decreased from 0.74 (3.50) to 0.20 (0.57) mg/kg/day ($p < 0.001$). Cystoid macular edema, which was present in 40 eyes at baseline, showed complete resolution in 28 eyes at 6 months. The mean suppression load decreased significantly (8.81 [5.05] versus 5.40 [4.43]; $p < 0.001$). Six months after the initiation of the study, 111 patients (85 %) were able to reduce at least 50 % of their baseline immunosuppression load. Only 9 patients (6.9 %) had severe relapses during the 6 months of follow-up. The authors concluded that adalimumab seems to be well-tolerated and helpful in decreasing inflammatory activity in refractory uveitis and may reduce steroid requirement. Moreover, they stated that further controlled studies of adalimumab for uveitis are needed.

In a case-cohort interventional study, Sen et al (2012) examined the use of adalimumab in the treatment of refractory non-infectious childhood chronic uveitis. This study was performed on patients with uveitis, who were treated with adalimumab after failure of treatment with a combination of corticosteroids and another immunosuppressant drug. Main outcome measures were

Adverse events and reasons for stopping adalimumab were noted. A total of 17 patients from a single regional center were included in the study; 9 patients had previously received an anti-TNF agent, and because of inefficacy, all were changed to adalimumab. At 12 months, fewer patients had visual acuity worse than LogMAR 0.4 (18 % versus 32 % at baseline). Using standardized uveitis nomenclature criteria, at 3 months, 50 % of the patients eyes ($n = 32$) had improved, 16 % had stable inflammation and 3 % had worsened,

whereas 31 % were maintained with no anterior chamber cells. Six patients required courses of oral steroids for uveitis. Seven patients received intra- or peri-ocular injections of steroids. Adalimumab treatment was interrupted in 1 patient because of varicella zoster infection. It was stopped in 3 patients; 7 (41 %) patients reported injection site reactions. The authors concluded that in this group of children with refractory uveitis, use of adalimumab was associated with improvement in visual acuity and improving or stable ocular inflammation. However, it did not completely obviate the need for systemic or peri-ocular steroid treatment. They stated that prospective randomized controlled trials are needed to help determine which subset of patients may benefit from adalimumab and the duration of treatment.

Bravo-Ljubetic et al. (2013) reported the results of adalimumab therapy in a cohort of children with refractory non-infectious uveitis. The medical records of patients diagnosed with uveitis and treated with bi-weekly adalimumab injections for a period of at least 3 months at the University Hospital of La Paz from 2007 to 2012 were retrospectively reviewed. Improvement in inflammatory activity was graded according to grading schema of the Standardization of Uveitis Nomenclature Working Group. A total of 15 patients participated in the study (12 girls; mean patient age of 12 years). Diagnoses were JIA in 10 patients, idiopathic uveitis in 4, and familial juvenile systemic granulomatosis or Blau syndrome in 1. Mean follow-up was 32 months (median of 36; range of 15 to 58 months). Improvement in inflammatory activity was initially observed in 12 (86 %) of 14 children, with a mean time to achieve response of 6 weeks (median of 4; range of 1 to 18). Treatment was effective in 9 patients (60 %), mildly effective in 2 (13 %), ineffective in 2 (13 %), and resulted in worsening in 2 (13 %). In the JIA patients, response was effective in 6 cases (60 %), mildly effective in 2 (20 %), and ineffective in 2 (20 %). Adalimumab therapy was discontinued in 4 patients. The authors concluded that adalimumab was effective in most patients in the initial control of acute inflammatory activity in children with refractory uveitis, although therapy appears to become less effective in the long-term. The main drawbacks of this study were its small sample size, wide age-group range, and lack of long-term follow-up.

In an open-label, comparative, multi-center, cohort study of childhood chronic uveitis, Simonini et al (2013) compared the effectiveness of adalimumab when used as first anti-TNF α therapy versus adalimumab used after the failure of a previous anti-TNF α (infliximab). A total of 26 patients (14 females, 12 males; median age of 8.6 years) with refractory, non-infectious active uveitis were enrolled. Due to the refractory course of uveitis to previous DMARD treatment, group 1 received adalimumab (24 mg/sq mt, every 2 weeks), as first anti-TNF α choice; group 2 received adalimumab, as second anti-TNF α drug, due to the loss of efficacy of infliximab, administered after a period of at least 1 year. Both groups received adalimumab for at least 1 year of treatment. Primary outcome was, once remission was achieved, the time to a first relapse. A total of 14 children (10 with JIA, 3 with idiopathic uveitis, 1 with Behçet's disease) were recruited in group 1; 12 children (7 with JIA, 3 with idiopathic uveitis, 1 with early-onset sarcoidosis, 1 with Behçet's disease) in group 2. Group 2 showed a lower probability to steroid discontinuation during the first 12 months of treatment (Mantel-Cox $\chi^2(2)4.12$, $p < 0.04$). In long-term follow-up, group 1 had higher probability of uveitis remission during the time of treatment on adalimumab (median $\pm$ SE: 18 ± 1.1 versus 4 ± 0.6 months, 95 % CI: 15.6 to 27.5 versus 2.7 to 5.2, Mantel-Cox $\chi^2(2)10.12$, $p < 0.002$). The authors concluded that even if limited to a relatively small group, the findings of this study suggested a better efficacy of adalimumab when used as first anti-TNF α treatment in childhood chronic uveitis. They stated that "further studies in a large cohort, in a prospective fashion, preferably by a randomized clinical trial, focused on one disease entity with a sufficient sample size, seem to be advocated to address this point".

A consensus panel organized by the American Uveitis Society (Levy-Clarke, et al., 2014) provided recommendations for the use of TNF- α biologic agents in patients with ocular inflammatory disorders. The authors performed a systematic review of literature to generate guidelines for use of these agents in ocular inflammatory conditions. Recommendations were generated using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) group criteria. The panel found that numerous studies including controlled clinical trials have demonstrated that anti-TNF- α biologic agents (in particular infliximab and adalimumab) are effective in the treatment of severe ocular inflammatory disease. Based on these studies, the panel made the following recommendations. Infliximab and adalimumab can be considered as first-line immunomodulatory agents for the treatment of ocular manifestations of Behçet's disease. Infliximab and adalimumab can be considered as second-line immunomodulatory agents for the treatment of uveitis associated with juvenile arthritis. Infliximab and adalimumab

can be considered as potential second-line immunomodulatory agents for the treatment of severe ocular inflammatory conditions including posterior uveitis, panuveitis, severe uveitis associated with seronegative spondyloarthritis, and scleritis in patients requiring immunomodulation in patients who have failed or who are not candidates for antimetabolite or calcineurin inhibitor immunomodulation. Infliximab and adalimumab can be considered in these patients in preference to etanercept, which seems to be associated with lower rates of treatment success.

Vogt-Koyanagi-Harada Syndrome

Rodríguez Rivera et al. (2011) noted that Vogt-Koyanagi-Harada syndrome is an autoimmune multi-system disease, characterized by the association of ocular inflammatory manifestations (uveitis and retinal detachment) and extra-ocular lesions (e.g., meningismus and tegumentary or auditory findings). These investigators reported the case of a Hispanic woman with this syndrome treated with adalimumab.

Jeroudi et al. (2014) stated that pediatric Vogt-Koyanagi-Harada syndrome (VKH) is rare, with limited cases of corticosteroid-sparing immunosuppression use reported. A 15-year old Hispanic girl was referred for bilateral intra-ocular inflammation. Her initial best corrected visual acuity (BCVA) was 20/30 in the right eye and 20/200 in the left eye, with granulomatous keratic precipitates, anterior chamber and vitreous cell, optic disc edema, and nummular depigmented chorio-retinal lesions on examination consistent with VKH after an unrevealing work-up. Inflammation was recurrent despite oral prednisone and methotrexate. Adalimumab led to rapid resolution of inflammation, successful dose reduction of prednisone and methotrexate, and final BCVA of 20/25 in the right eye and 20/40 in the left at 26-month follow-up.

An UpToDate review on "Uveitis: Treatment" (Rosenbaum, 2014) states that "Vogt-Koyanagi-Harada disease may benefit from intravenous methylprednisolone when the illness first begins The role of TNF-alpha inhibitors such as infliximab, adalimumab, or etanercept in the management of patients with uveitis is uncertain".

Similarly, a review on "Vogt-Koyanagi-Harada Disease" (Walton, 2014) stated that "Case reports suggest that intravenous immunoglobulins (IVIGs) and infliximab may be of interest in the treatment for VKH syndrome. Further trials are needed to assess the efficacy of these agents".

Other Indications

Ramos-Casals et al. (2008) stated that in 2006, the Study Group on Autoimmune Diseases (GEAS) of the Spanish Society of Internal Medicine created the BIOGEAS project, a multi-center study devoted to collecting data on the use of biological agents in adult patients with systemic autoimmune diseases (SAD). The information source is a periodic surveillance of reported cases by a MEDLINE search (last update before this writing: December 31, 2007). The analysis included a total of 19 SAD and 6 biological agents. By December 31, 2007, the Registry included 1,370 patients with SAD who had been treated with biological agents (562 received infliximab, 463 rituximab, 285 etanercept, 42 anakinra, and 18 adalimumab). Systemic autoimmune diseases included Sjogren syndrome (SS; 215 cases), Wegener granulomatosis (261 cases), sarcoidosis (219 cases), systemic lupus erythematosus (SLE; 172 cases), Behçet disease (173 cases), adult-onset Still disease (118 cases), cryoglobulinemia (88 cases), and other diseases (80 cases). The higher rate of therapeutic response was found for the use of rituximab in patients with SLE (90 %), SS (91 %), anti-phospholipid syndrome (92 %), and cryoglobulinemia (87 %); infliximab in sarcoidosis (99 %), adult-onset Still disease (90 %), and polyarthritides (86 %); and etanercept in Behçet disease (96 %). Results from controlled trials showed lack of efficacy for the use of infliximab in SS and etanercept in SS, Wegener granulomatosis, and sarcoidosis. In addition, an excess of side effects (greater than 50 % of reported cases) was observed for the use of infliximab in inflammatory myopathies and sarcoidosis, and for the use of etanercept in polymyositis. Sufficient data are not yet available to evaluate fully the safety and effectiveness of adalimumab and anakinra in patients with SAD. The authors concluded that current scientific evidence on the use of biological therapies in patients with SAD is mainly based on uncontrolled, observational data. The best results have been observed in the use of rituximab for SS, SLE, and cryoglobulinemia; infliximab for sarcoidosis and adult-onset Still disease; and etanercept for Behçet disease. Lack of efficacy was demonstrated for infliximab and etanercept in SS, for etanercept in Wegener granulomatosis and sarcoidosis, and for TNF in SS. They stated that future controlled trials are needed to confirm the potential use

of biological therapies in patients with SAD.

Adalimumab Serum Levels and Antibodies

For measurement of serum levels of adalimumab and/or antibodies to adalimumab, see [CPB 1091 - Therapeutic Drug Monitoring of Anti-Tumor Necrosis Factor Blocking Agents](#).

Appendix

Appendix A: Examples of Clinical Reasons to Avoid Pharmacologic Treatment with Methotrexate, Hydroxychloroquine, Cyclosporine, Acitretin, or Leflunomide

- Clinical diagnosis of alcohol use disorder, alcoholic liver disease or other chronic liver disease
- Drug interaction
- Risk of treatment-related toxicity
- Pregnancy or currently planning pregnancy
- Breastfeeding
- Significant comorbidity prohibits use of systemic agents (e.g., liver or kidney disease, blood dyscrasias, uncontrolled hypertension)
- Hypersensitivity
- History of intolerance or adverse event

Appendix B: Risk factors for Articular Juvenile Idiopathic Arthritis

1. Positive rheumatoid factor
2. Positive anti-cyclic citrullinated peptide antibodies
3. Pre-existing joint damage

Table: Hurley Staging System

Stage	Characteristics
I	Solitary or multiple isolated abscess formation without scarring or sinus tracts. (A few minor sites with rare inflammation; may be mistaken for acne.)
II	Recurrent abscesses, single or multiple widely separated lesions, with sinus tract formation. (Frequent inflammation restrict movement and may require minor surgery such as incision and drainage.)
III	Diffuse or broad involvement across a regional area with multiple interconnected sinus tracts and abscesses. (Inflammation of sites to the size of golf balls, or sometimes baseballs; scarring develops, including subcutaneous tracts of infection – see fistula. Obviously, patients at this stage may be unable to function.)

Source: Wiecezorek and Walecka (2018)

Table: Brands of Targeted Immune Modulators and FDA-approved Indications (not an all-inclusive list)

Brand Name	Generic Name	FDA Labeled Indications
Abrilada	adalimumab-afzb	Ankylosing spondylitis and non-radiographic axial spondyloarthritis Crohn's disease Hidradenitis suppurativa Plaque psoriasis Psoriatic arthritis

		Rheumatoid arthritis Ulcerative colitis Uveitis
Actemra	tocilizumab	COVID-19 in hospitalized adults Cytokine release syndrome (CRS) Giant cell arteritis Juvenile idiopathic arthritis Rheumatoid arthritis Systemic juvenile idiopathic arthritis Systemic sclerosis-associated interstitial lung disease (SSc-ILD)
Amjevita	adalimumab-atto	Ankylosing spondylitis Crohn's disease Hidradenitis suppurativa Juvenile idiopathic arthritis Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis Uveitis
Arcalyst	rilonacept	Cryopyrin-associated periodic syndromes Deficiency of interleukin-1 receptor antagonist (DIRA) Recurrent pericarditis
Cimzia	certolizumab	Ankylosing spondylitis or non-radiographic axial spondyloarthritis Crohn's disease Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis
Cosentyx	secukinumab	Ankylosing spondylitis or non-radiographic axial spondyloarthritis Enthesitis-related arthritis Hidradenitis suppurativa Plaque psoriasis Psoriatic arthritis
Cyltezo	adalimumab-adbm	Ankylosing spondylitis Crohn's disease Hidradenitis suppurativa Juvenile idiopathic arthritis Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis Uveitis
Enbrel	etanercept	Ankylosing spondylitis Juvenile idiopathic arthritis Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis
Entyvio	vedolizumab	Crohn's disease Ulcerative colitis

Hadlima	adalimumab-bwwd	Ankylosing spondylitis Crohn's disease Hidradenitis suppurativa Juvenile idiopathic arthritis Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis Uveitis
Hulio	adalimumab-fkjp	Ankylosing spondylitis Crohn's disease Hidradenitis suppurativa Juvenile idiopathic arthritis Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis Uveitis
Humira	adalimumab	Ankylosing spondylitis Crohn's disease Hidradenitis suppurativa Juvenile idiopathic arthritis Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis Uveitis
Hyrimoz	adalimumab-adaz	Ankylosing spondylitis Crohn's disease Hidradenitis suppurativa Juvenile idiopathic arthritis Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis Uveitis
Idacio	adalimumab-aacf	Ankylosing spondylitis Crohn's disease Hidradenitis suppurativa Juvenile idiopathic arthritis Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis Uveitis
Ilaris	canakinumab	Adult-onset Still's disease Gout flares Periodic fever syndromes Systemic juvenile idiopathic arthritis
Ilumya	tildrakizumab-asmn	Plaque psoriasis
Kevzara	sarilumab	Juvenile idiopathic arthritis Polymyalgia rheumatica Rheumatoid arthritis

Kineret	anakinra	Cryopyrin-associated periodic syndromes Deficiency of interleukin-1 receptor antagonist (DIRA) Rheumatoid arthritis
Olumiant	baricitinib	Alopecia areata COVID-19 in hospitalized adults Rheumatoid arthritis
Orencia	abatacept	Acute graft versus host disease Juvenile idiopathic arthritis Psoriatic arthritis Rheumatoid arthritis
Otezla	apremilast	Oral ulcers associated with Behçet's Disease Plaque psoriasis Psoriatic arthritis
Remicade (for Remicade biosimilars, see CPB 0341 - Infliximab)	infliximab	Ankylosing spondylitis Crohn's disease Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis
Rinvoq	upadacitinib	Ankylosing spondylitis or non-radiographic axial spondyloarthritis Atopic dermatitis Crohn's disease Juvenile idiopathic arthritis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis
Rituxan (for Rituxan biosimilars, see CPB 0314 - Rituximab)	rituximab	Chronic lymphocytic leukemia Granulomatosis with polyangiitis Microscopic polyangiitis Pemphigus vulgaris Rheumatoid arthritis Various subtypes of non-Hodgkin's lymphoma
Siliq	brodalumab	Plaque psoriasis
Simponi	golimumab	Ankylosing spondylitis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis
Simponi Aria	golimumab intravenous	Ankylosing spondylitis Juvenile idiopathic arthritis Psoriatic arthritis Rheumatoid arthritis
Skyrizi	risankizumab-rzaa	Crohn's disease Plaque psoriasis Psoriatic arthritis Ulcerative colitis
Stelara	ustekinumab	Crohn's disease Plaque psoriasis Psoriatic arthritis Ulcerative colitis

Taltz	ixekinumab	Ankylosing spondylitis or non-radiographic axial spondyloarthritis Plaque psoriasis Psoriatic arthritis
Tremfya	guselkumab	Plaque psoriasis Psoriatic arthritis Ulcerative colitis
Tysabri	natalizumab	Crohn's disease Multiple sclerosis
Xeljanz	tofacitinib	Ankylosing Spondylitis Polyarticular Course Juvenile Idiopathic Arthritis Psoriatic arthritis Rheumatoid arthritis Ulcerative Colitis
Xeljanz XR	tofacitinib, extended release	Ankylosing Spondylitis Polyarticular Course Juvenile Idiopathic Arthritis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis
Yuflyma	adalimumab-aaty	Ankylosing spondylitis Crohn's disease Hidradenitis Suppurativa Juvenile idiopathic arthritis Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis Uveitis
Yusimry	adalimumab-aqvh	Ankylosing spondylitis Crohn's disease Hidradenitis suppurativa Juvenile idiopathic arthritis Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis Uveitis

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